Hyperspectral Anomaly Detection With Guided Autoencoder

Pei Xiang, Shahzad Ali, Soon Ki Jung, *Senior Member, IEEE*, and Huixin Zhou, *Member, IEEE*

***Abstract*— Recently, autoencoder (AE)-based hyperspectral anomaly detection methods have demonstrated excellent performance on hyperspectral images (HSIs). The AE can simultaneously reconstruct both the anomaly targets and the background, but the lack of prior information limits the ability to detect anomalies. This study proposes a novel hyperspectral anomaly detection method based on a guided AE to reduce the feature representation for the anomaly targets. First, a multilayer AE network with skip connections is proposed to fully extract the abundant latent features from HSIs and enhance the expressive ability of the network. The reconstructed HSI can be obtained by the proposed AE network. Second, to suppress the anomaly targets in the obtained reconstructed HSI and better represent background features, a guided module based on a guided image is added to the network to reduce the feature representation of the anomaly targets by providing feedback information. Moreover, the guided image is calculated using a proposed spectral similarity method that uses the local spatial features of the HSI. Finally, we use the reconstruction error as a performance metric and compare the results of our proposed method with other state-of-the-art methods on six real-world HSIs. The results demonstrate the effectiveness and superiority of the proposed method.**

***Index Terms*— Anomaly detection, autoencoder (AE), guide image, hyperspectral image (HSI).**

I. Introduction

HYPERSPECTRAL image (HSI) is a 3-D data cube acquired by a hyperspectral imager that combines imaging and spectroscopy techniques [1], [2]. An HSI supplies 2-D spatial information of objects along with detailed spectral information [3]. Such fine spectral information can reflect the essential properties of the objects. The combination of images and spectra gives HSIs a huge advantage in target detection [4].

Manuscript received 12 May 2022; revised 18 August 2022; accepted 10 September 2022. Date of publication 16 September 2022; date of current version 3 October 2022. This work was supported in part by the Ministry of Education, School of Computer Science and Engineering, Kyungpook National University, South Korea, through the BK21 FOUR Project (AI-Driven Convergence Software Education Research Program), under Grant 4199990214394; in part by the Overseas Expertise Introduction Project for Discipline Innovation under Grant B17035; and in part by the China Scholarship Council for Joint Ph.D. Students under Grant 202106960025. *(Corresponding author: Soon Ki Jung.)*

Pei Xiang is with the School of Physics and Optoelectronic Engineering, Xidian University, Xi'an 710071, China, and also with the School of Computer Science and Engineering, Kyungpook National University, Daegu 41566, Republic of Korea (e-mail: pxiang@stu.xidian.edu.cn).

Shahzad Ali and Soon Ki Jung are with the School of Computer Science and Engineering, Kyungpook National University, Daegu 41566, Republic of Korea (e-mail: shahzadali@knu.ac.kr; skjung@knu.ac.kr).

Huixin Zhou is with the School of Physics and Optoelectronic Engineering, Xidian University, Xi'an 710071, China (e-mail: hxzhou@mail.xidian.edu.cn).

Digital Object Identifier 10.1109/TGRS.2022.3207165

Hyperspectral target detection is divided into target detection with prior spectral information and anomaly detection without prior spectral information [5]. Acquiring the spectral information of the targets and the background in advance is challenging. Therefore, hyperspectral anomaly detection does not rely on such priors, which has a significant practical application value. Without this prior information, the anomaly targets are detected by calculating the difference between the HSI pixels [6]. At present, hyperspectral anomaly detection is being widely used in smart agriculture, mineral exploration, environmental supervision, and search and rescue operations [7], [8].

Every year, various anomaly detection methods are proposed. These methods can be classified into three groups based on their structure and theoretical basis: probability-distribution-based, representation-based, and deep-learning-based [9]. The probability-distribution-based anomaly detection method generally models the background of the HSI, and pixels that do not obey this model are considered the anomaly targets. However, due to the complex distribution of the actual background, it is difficult for a simple model to describe the background distribution, which limits the detection performance [10]. The representation-based method reconstructs each pixel in the HSI by the method of representation. The reconstruction error is used as anomaly detection result. Although this method has achieved satisfactory results, it is difficult to construct an overcomplete dictionary in the representation. There are many parameters set manually [11]. Different from the above two methods, the deep-learning-based method enhances the discrimination between the background and anomaly targets by exploiting the latent high-level features in the HSI. In addition, due to the lack of spectral information of the background and anomaly targets in anomaly detection, the unsupervised deep learning methods have received extensive attention in anomaly detection. Autoencoder (AE), as an unsupervised learning method in deep learning, has been successfully applied in hyperspectral anomaly detection and achieved excellent detection results [12].

In hyperspectral anomaly detection, the anomaly targets have only a few pixels, and most of the pixels are background. When all the pixels are input into the AE, it can be considered that the AE mainly learns the features of the background. Then, its ability to reconstruct the anomaly target pixels is not as good as that of the background pixels [13], [14]. The reconstruction error of the background pixels is small, while the reconstruction error of the anomaly target pixels is large. Therefore, the AE-based anomaly detection

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

method uses this reconstruction error as the final anomaly score.

Although AE effectively improves the anomaly detection performance, due to the lack of prior spectral information of the background and anomaly targets, the entire HSI is usually used as the training set in the training of AE. It has been proven that AE can successfully reconstruct the anomaly targets even if they only occupy very few pixels [15], [16]. This can have an impact on the anomaly detection performance. In addition, when using AE to reconstruct each pixel in the HSI, the correlation between the pixels and the spatial features of the HSI is not considered.

To solve the above problems, this study proposes a novel hyperspectral anomaly detection method based on a guided AE. First, a multilayer AE network with skip connections is proposed, which extracts rich latent HSI features and improves the feature expression ability of the AE network. Second, to reduce the reconstruction of the anomaly targets, a guided module based on the guided image is introduced during the backpropagation of the network. We obtain this guided image by further proposing a spectral similarity method, which fully exploits the local spatial features of the HSI. The AE reconstructs an HSI with weakened anomaly targets through the guided module. Finally, the reconstruction error of the guided AE is used to measure the anomaly detection performance. Our contributions are threefold.

1) We propose a guided AE framework for hyperspectral anomaly detection by exploiting the rich latent features of an HSI. The guided module embedded in the AE network constrains its feature representation ability for the anomaly targets.
2) We propose a spectral-similarity-based method to obtain the guided image by taking advantage of the local spatial features of an HSI.
3) We demonstrate the effectiveness of our approach with an extensive and detailed experimental result on six datasets, including a comparison with the state-of-the-art methods, as well as an in-depth analysis of our approach with a detailed ablation study on the effects of noise and smaller anomaly targets.

The rest of this study is organized as follows. Section II summarizes the related work. Section III describes the proposed method in detail. Section IV presents and analyzes the experimental results. Section V summarizes the study.

II. Related Work

A. Probability-Distribution-Based Anomaly Detection

RX, proposed by Reed and Xiaoli in 1990, is the most traditional probability-distribution-based approach to anomaly detection [17]. It detected the anomaly targets using the Mahalanobis distance pixel by pixel under the assumptions that the background obeys a multivariate Gaussian distribution. The RX method is further divided into global RX (GRX) methods [18] and local RX (LRX) methods [19]. To evaluate the background statistics, the GRX method uses global background, while the LRX method uses local background. However, the anomaly targets and noise can easily disturb this process, and the assumption about background following a multivariate Gaussian distribution fails to describe a complex background. As a result, a variety of improved methods have been developed. The subspace-based RX (SSRX) method proposed by Schaum [20] extracted the first few principal components of the covariance matrix in the RX method to better represent the background statistics. The weighted RX (WRX) method developed by Guo *et al.* [21] improved the detection performance by adding different pixel weights to the anomaly targets and background. The component decomposition analysis (CDA) proposed by Chen *et al.* [22] decomposed an HSI into a linear sum of principal, independent, and noise components to better extract the anomaly targets. Kwon and Nasrabadi [23] proposed the kernel RX (KRX) method to project an original HSI into a high-dimensional space separating the anomaly targets and background and solving the linear inseparability problem in the original HSI.

B. Representation-Based Anomaly Detection

With the development of compressed sensing theory, the representation-based anomaly detection methods have been widely applied to HSIs. It was Chen *et al.* [24] who introduced sparse representation into hyperspectral target detection for the first time. The collaborative-representation-based detector (CRD) proposed by Li and Du [25] considered every pixel from the outer window to represent a center pixel to be tested. Combined with low-rank and sparse properties in an HSI, robust principal component analysis (RPCA) [26] decomposed it into the low-rank and sparse matrices. However, the RPCA projected an HSI into a subspace, which was susceptible to noise. To reduce the influence of noise on the detection results, the low-rank and sparse matrix decomposition (LRaSMD) method developed by Sun *et al.* [27] decomposed an HSI into a low-rank matrix, a sparse matrix, and a noise matrix. To exploit the background information in the LRaSMD method, the LRaSMD-based Mahalanobis distance method for anomaly detection (LSMAD) was proposed by Zhang *et al.* [28]. Xu *et al.* proposed the low-rank and sparse representation (LRASR) method [29] for anomaly detection using low-rank representation (LRR). The LRASR method adopted the constructed background dictionary to obtain background representation coefficients and a sparse matrix. To address the mixed pixel problem, Qu *et al.* [30] introduced spectral unmixing for anomaly detection and proposed an abundance and dictionary-based low-rank decomposition (ADLR) method. By introducing a mixture noise model into the sparse matrix, the low-rank and sparse decomposition with mixture of Gaussian (LSDM-MoG) method was devised by Li *et al.* [31] to improve the accuracy of anomaly decomposition. Besides, the prior-based tensor approximation (PTA) method [32] and the graph and total variation regularized low-rank representation (GTVLRR) method [33] explored the tensor structure and the low-rank manifold structure of an HSI, respectively.

C. AE-Based Anomaly Detection

Because of the excellent performance of the AE, it has been introduced into hyperspectral anomaly detection.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

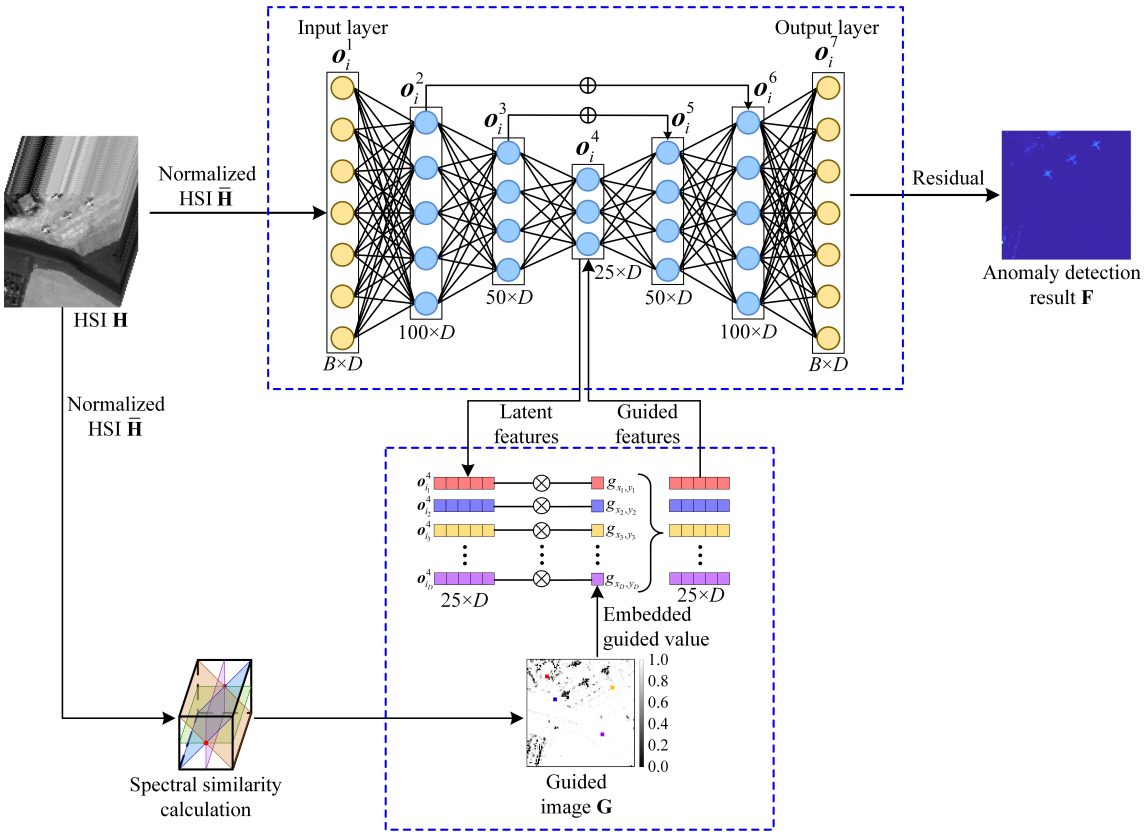

Fig. 1. Overall pipeline of the proposed guided AE method.

Chang *et al.* [34] proposed an anomaly detection method based on a sparse AE, which took the difference between the reconstruction errors of the pixels from the whole window and the pixels from the outer window as the detection result in the dual-window model. Xie *et al.* [35] proposed the spectral constraint adversarial AE to fully exploit the hidden layer features and suppress the background. Zhao and Zhang [36] introduced the spectral–spatial stacked AE, which first preprocessed an HSI by LRaSMD. These matrices were reconstructed by AE to obtain a feature matrix for anomaly target extraction. The autoencoding adversarial networks proposed by Arisoy *et al.* [37] first preprocessed the data for background purification. Then, an HSI was reconstructed to exploit spectral and spatial information through three different deep autoencoding adversarial networks. Lu *et al.* [38] proposed a manifold-constrained AE network to exploit local features in the learning process by embedding the manifold learning method. To reduce the influence of anomaly on the detection results, Xie *et al.* [14] proposed a method based on the autoencoding adversarial networks. The method first applied background samples extracted from an input HSI to train an AE. Then, it applied the trained model to an entire HSI to increase the difference between the anomaly targets and the background in the reconstruction error. Robust graph AE (RGAE) was developed by Fan *et al.* [15] by adding graph regularization to improve the robustness of the network against the anomaly targets and noise. Using fully convolutional AE, an automatic hyperspectral anomaly detection method called Auto-AD was proposed by Wang *et al.* [16]. This method added an adaptive-weighted loss function to the feedback of the fully convolutional network reducing the reconstruction of the anomaly targets.

The above methods used different approaches to reduce the influence of anomaly targets over the AE reconstruction errors and improve the detection probability. However, these methods may still be disturbed by the anomaly targets and noise during an HSI reconstruction affecting the detection results. Different from these methods, we proposed a guided AE network to reduce the representation ability of an AE for the anomaly targets through a guided module. We proposed a spectral similarity method that combines the spatial and spectral features of an HSI forming a guided image. At the same time, the expressive ability of our network is improved using multilayer skip connections.

III. Proposed Approach

The framework for the proposed guided AE-based hyperspectral anomaly detector is shown in Fig. 1. The proposed method is divided into three steps. First, the HSI is reconstructed through a seven-layer AE network and skip connections. Second, combined with local spatial features, the guided image is obtained by calculating the spectral similarity. Finally, this guided image is added to the AE network, directing it to focus on reconstructing the background.

A. Multilayer AE With Skip Connections

Currently, some AE-based hyperspectral anomaly detection methods use a single hidden layer or a small number of hidden

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

layers to compress the feature dimension [39]. This limits the expressive ability of a network and further reduces its reconstruction ability. Hence, a seven-layer AE along with the skip connections is proposed to fully extract essential HSI features and improve the network representation capability.

The proposed AE network structure is shown in Fig. 1. We used fully connected layers to better use the spectral information from an HSI. Also, the features extracted from the second and third layers are combined with those extracted from the sixth and fifth layers, respectively, through skip connections, allowing the features of the different layers to be better used. The sigmoid function [40] serves as the activation function in the entire AE network.

We denoted an input HSI as $\mathbf{H} = (\boldsymbol{h}_1, \boldsymbol{h}_2, \ldots, \boldsymbol{h}_N) \in \mathbf{R}_{B\times N}$, where B is the number of bands and N is the total number of pixels in an HSI. Each pixel in an input HSI is regarded as a sample. Prior to feeding an HSI into the input layers, we normalized each pixel through min–max normalization. This step helps narrow down the spectral curves corresponding to the background pixels. The min–max normalization is performed as

$$\overline{\boldsymbol{h}}_i = \frac{1}{\boldsymbol{h}_{\max} - \boldsymbol{h}_{\min}} \odot (\boldsymbol{h}_i - \boldsymbol{h}_{\min}), \quad i = 1, 2, \ldots, N \tag{1}$$

where $\boldsymbol{h}_{\min}$ and $\boldsymbol{h}_{\max}$ represent the vector consisting of the minimum and maximum values of all the bands in an HSI, respectively, $\boldsymbol{h}_i$ is the ith pixel in the HSI $\mathbf{H}$, $\overline{\boldsymbol{h}}_i$ is the ith pixel after normalization, and $\odot$ is defined as the matrix dot product. The normalized HSI is denoted as $\overline{\mathbf{H}} = (\overline{\boldsymbol{h}}_1, \overline{\boldsymbol{h}}_2, \ldots, \overline{\boldsymbol{h}}_N) \in \mathbf{R}_{B\times N}$.

For the ith pixel $\overline{\boldsymbol{h}}_i$, the output of each layer after the fully connected layer and activation function is

$$\boldsymbol{z}_i^l = \mathbf{W}^l \boldsymbol{o}_i^{l-1} + \boldsymbol{b}^l \tag{2}$$

$$\boldsymbol{o}_i^l = \frac{1}{1 + \mathrm{e}^{-\boldsymbol{z}_i^l}} \tag{3}$$

where $\mathbf{W}^l$ and $\boldsymbol{b}^l$ are the weights and biases for a layer, respectively, $l \in \{1, 2, 3\}$ is the number of layers, and $\boldsymbol{o}_i^l$ is the output of each layer. The input to the network is the normalized pixels, as follows: $\boldsymbol{o}_i^1 = \overline{\boldsymbol{h}}_i$.

Because of skip connections, the outputs of the fifth and sixth layers are denoted as

$$\begin{cases} \boldsymbol{z}_i^6 &= \mathbf{W}^6\left(\boldsymbol{o}_i^5 + \boldsymbol{o}_i^3\right) + \boldsymbol{b}^6 \\ \boldsymbol{o}_i^6 &= \dfrac{1}{1 + \mathrm{e}^{-\boldsymbol{z}_i^6}} \end{cases} \tag{4}$$

$$\begin{cases} \boldsymbol{z}_i^7 &= \mathbf{W}^7\left(\boldsymbol{o}_i^6 + \boldsymbol{o}_i^2\right) + \boldsymbol{b}^7 \\ \boldsymbol{o}_i^7 &= \dfrac{1}{1 + \mathrm{e}^{-\boldsymbol{z}_i^7}} \end{cases} \tag{5}$$

where $\boldsymbol{o}_i^6$ and $\boldsymbol{o}_i^7$ are the outputs of the fifth and sixth layers, respectively.

The output $\boldsymbol{o}_i^7$ of the sixth layer is the final output of the network, which is used to calculate the reconstruction error and measure the performance of the proposed AE network.

B. Guided Image Computation Based on Spectral Similarity

A guided image is used as the basis for the guided module in the proposed guided AE network. An efficient method for generating guided images based on spectral similarity is presented in this section. The spectral similarity fully considers the relationship between pixels and local spatial features in an HSI.

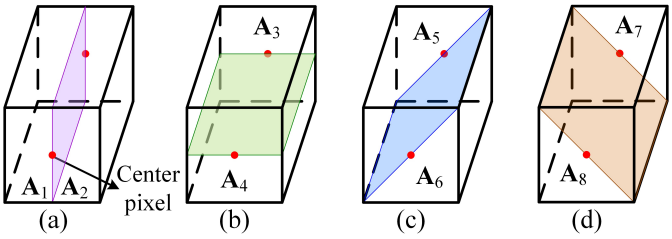

Fig. 2. Window is divided into eight subvolumes and the center pixel represents the pixel to be tested. (a) Subvolumes $\mathbf{A}_1$ and $\mathbf{A}_2$. (b) Subvolumes $\mathbf{A}_3$ and $\mathbf{A}_4$. (c) Subvolumes $\mathbf{A}_5$ and $\mathbf{A}_6$. (d) Subvolumes $\mathbf{A}_7$ and $\mathbf{A}_8$.

To exploit the local spatial features in the HSI, the classic hyperspectral anomaly detection approach exploits a dual-window model [41]. However, when this dual window is used for background statistical feature analysis, its size needs to be determined and the statistical feature may be disturbed by the anomaly targets and noise. Therefore, to further reduce the interference of the anomaly targets and noise in using local spatial features, inspired by the window model in curvature filters [42], we adopt a single-window model, which is internally divided into different subvolumes. A single window of size $c \times c$ is shown explicitly in Fig. 2. It can be divided into eight subvolumes by four segmentation methods. The eight subvolumes are denoted as $\mathbf{A}_j (j = 1, 2, \ldots, 8)$, and the center pixel at the middle is the pixel to be tested. For this window model, the following four cases exist.

1) When both the pixel to be tested and the pixels in the window belong to the background, the subvolumes with the highest similarity to the pixel to be tested can be easily selected using spectral similarity scores. There will be no interference from the anomaly targets and noise during the calculation process.
2) When the pixel to be tested is the background pixel and the anomaly targets' pixels are included in the window, the spectral similarity scores between the subvolumes containing the anomaly targets' pixels and the pixel to be tested are lower than the spectral similarity scores between other subvolumes and the pixel to be tested.
3) When the pixel to be tested is an anomaly target pixel and the pixels in the window are background pixels, the spectra of the pixels in all the subvolumes are quite different from the spectrum of the pixel to be tested. Then, the obtained spectral similarity scores will be relatively low.
4) When the pixel to be tested is an anomaly target pixel and the window contains anomaly targets' pixels, the spectral similarity scores between the subvolumes containing the anomaly targets pixels and the pixel to be tested must be higher than the spectral similarity scores between the subvolumes not containing the anomaly targets pixels and the pixel to be tested. However, because the anomaly targets only occupy a few pixels, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Calculation of Guided Image

Input: HSI **H**
Initialization: Window size $c \times c \times B$
For each pixel to be tested $\boldsymbol{t}_{x,y}$ **do**
1: Calculate the mean vector $\boldsymbol{u}_j$ of each subvolume in the window;
2: Calculate the difference value d_j according to (6);
3: Calculate the spectral similarity score between the pixel to be tested and each subvolume according to (7);
4: Choose the largest spectral similarity score as the value $g_{x,y}$ of the guided image **G** according to (8).
End For
Output: Guided image **G**

background pixels are still the majority in all the subvolumes. Hence, the spectral difference between each sub volume and the pixel to be tested is relatively high, and the obtained spectral similarity score is relatively low.

The spectral similarity scores for the background pixels will be high, whereas the anomaly targets will have low scores. Finally, the obtained similarity scores for all the pixels are used to calculate the guide image **G**, as follows.

First, the mean vector $\boldsymbol{u}_j$ of each subvolume in the window is calculated. Second, the difference value d_j between the mean vector $\boldsymbol{u}_j$ of each subvolume and the pixel to be tested is obtained by

$$d_j = \sum_{b=1}^{B} \left[\boldsymbol{t}_{x,y}(b) - \boldsymbol{u}_j(b)\right]^2 \tag{6}$$

where $\boldsymbol{t}_{x,y}(b)$ represents the value of the bth element in the pixel to be tested, $\boldsymbol{t}_{x,y}$, in the HSI; x and y represent the 2-D spatial coordinates of the pixel to be tested in the HSI, and $x \times y = N$; and $\boldsymbol{u}_j(b)$ represents the value of the bth element in the mean vector $\boldsymbol{u}_j$. The spectral similarity score between the pixel to be tested and each subvolume is

$$s_{x,y}^{j} = e^{-d_j^2} \tag{7}$$

where $s_{x,y}^{j}$ is the spectral similarity score between the subvolume $\mathbf{A}_j$ and the pixel to be tested, $\boldsymbol{t}_{x,y}$. Finally, the maximum value among the eight spectral similarity scores is selected as the similarity score of the pixel to be tested. The selection process can be written as

$$g_{x,y} = \max_j \left(s_{x,y}^{j}\right) \tag{8}$$

where $g_{x,y}$ is the value at (x, y) in the guide image **G**, and $\max(\cdot)$ indicates the operation of selecting the maximum value. The obtained similarity scores of all the pixels are used as the guided image. It is worth noting that when using the proposed window model to calculate the value of the guided image for the image border, the values within the window outside the image are padded by mirror reflection of the image border elements. The detailed calculation steps of the guided image are shown in Algorithm 1.

C. Guided Module

An AE network usually uses an entire HSI as input data. The AE network still reconstructs anomalies in an HSI despite only occupying a few pixels [43], [44]. During the backpropagation of the AE network, the proposed guided module is used to further strengthen the representation of the background and reduce the reconstruction of the anomaly targets. As a result, our guided AE can reconstruct a quality HSI without anomaly targets.

Combined with the extracted latent features, the guided module is embedded into the middle hidden layer of the network. The value of the guided image calculated at the corresponding position in the HSI sample assigns a weight to each extracted feature by the network. The detailed embedding is written as

$$\widetilde{\boldsymbol{o}}_i^4 = \beta g_{x,y} \boldsymbol{o}_i^4 \tag{9}$$

where $\boldsymbol{o}_i^4$ is the feature of a sample $\overline{\boldsymbol{h}}_i$ extracted by the network, the 2-D spatial position of sample $\overline{\boldsymbol{h}}_i$ in the HSI is (x, y), $g_{x,y}$ is the value of the guide image **G** computed at (x, y) in the HSI **H**, $\widetilde{\boldsymbol{o}}_i^4$ is the feature of the sample $\overline{\boldsymbol{h}}_i$ after applying the guided module, and β is the penalty coefficient. This guides the network to pay more attention to the background features and reduce the influence of anomaly target features.

As shown in Fig. 1, D pixels $(\overline{\boldsymbol{h}}_{x_1,y_1}, \overline{\boldsymbol{h}}_{x_2,y_2}, \ldots, \overline{\boldsymbol{h}}_{x_D,y_D})$ in an HSI are used for training where D represents the number of selected pixels and (x, y) represents the spatial coordinates. In the guided module, the input latent feature values $(\boldsymbol{o}_{i_1}^4, \boldsymbol{o}_{i_2}^4, \ldots, \boldsymbol{o}_{i_D}^4)$ are multiplied by the corresponding guided values $(g_{x_1,y_1}, g_{x_2,y_2}, \ldots, g_{x_D,y_D})$ to form the guided latent features $(\widetilde{\boldsymbol{o}}_{i_1}^4, \widetilde{\boldsymbol{o}}_{i_2}^4, \ldots, \widetilde{\boldsymbol{o}}_{i_D}^4)$. These guided latent features then replace the original latent features of the network. As a result, the guided module further improves the feature representation of the background, thus reducing the reconstruction of the anomalous targets.

This study uses the traditional backpropagation algorithm [45] to train the proposed guided AE network. The mean square error (MSE) is used as the loss function. The MSE is defined as

$$\text{loss} = \frac{1}{2N} \sum_{i=1}^{N} \sum_{b=1}^{B} \left[\left(\overline{\boldsymbol{h}}_i(b) - \boldsymbol{o}_i^7(b)\right)\right]^2 \tag{10}$$

where $\overline{\boldsymbol{h}}_i(b)$ is the bth element in the ith pixel $\overline{\boldsymbol{h}}_i$, and $\boldsymbol{o}_i^7(b)$ is the bth element in the output $\boldsymbol{o}_i^7$ of the pixel $\overline{\boldsymbol{h}}_i$. During training, the learning rate is l_{a}. A maximum iteration value r_{iter} is used as a stopping condition for training.

D. Extraction of Anomaly Targets

Once the guided AE network is fully trained, the HSI $\widetilde{\mathbf{H}} = (\widetilde{\boldsymbol{h}}_1, \widetilde{\boldsymbol{h}}_2, \ldots, \widetilde{\boldsymbol{h}}_N) = (\boldsymbol{o}_1^7, \boldsymbol{o}_2^7, \ldots, \boldsymbol{o}_N^7)$ is reconstructed at the test time. We take the residual between the input HSI and the reconstructed HSI as the anomaly detection result. Then, the anomaly detection result is represented as

$$f_i = \sum_{b=1}^{B} \left[\overline{\boldsymbol{h}}_i(b) - \widetilde{\boldsymbol{h}}_i(b)\right]^2 \tag{11}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

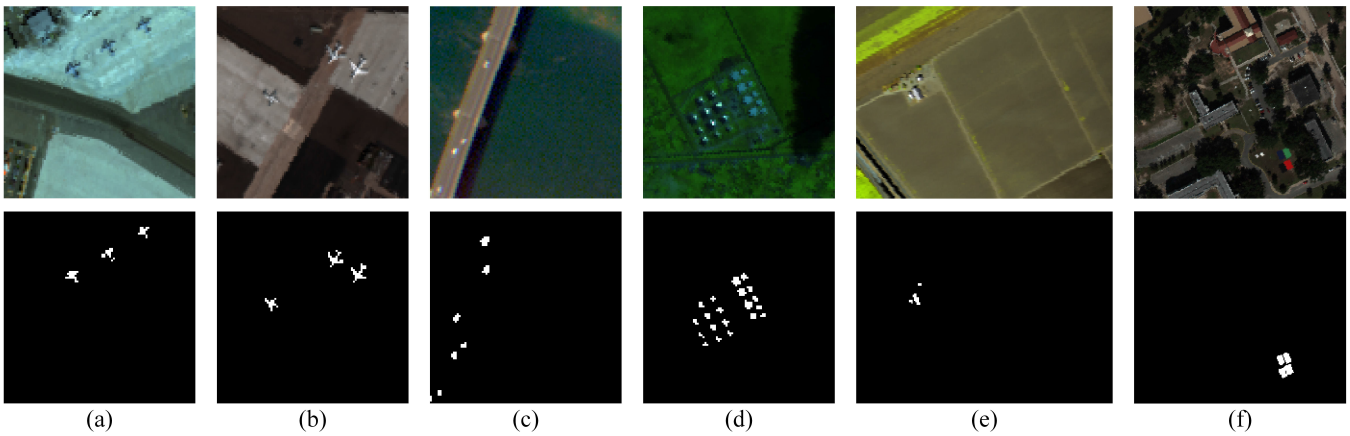

Fig. 3. False-color images and ground-truth images of the experimental datasets. (a) San Diego I. (b) San Diego II. (c) Pavia. (d) Texas Coast. (e) Salinas. (f) MUUFL Gulfport.

where $\widetilde{\boldsymbol{h}}_i(b)(i = 1, 2, \ldots, N)$ is the bth element in the reconstructed hyperspectral pixel $\widetilde{\boldsymbol{h}}_i$, $\overline{\boldsymbol{h}}_i(b)$ is the bth element in the input pixel $\overline{\boldsymbol{h}}_i$, and f_i represents the value of the ith pixel of the anomaly detection result $\mathbf{F}$, where $\mathbf{F} = \{f_i\}$. The larger the value of f_i, the more likely the pixel will be an anomaly target. The detailed steps of the proposed guided AE are introduced in Algorithm 2.

Algorithm 2 Proposed Guided AE Method

Input: HSI $\mathbf{H}$

Initialization: Window size c, learning rate l_a, penalty coefficient β, number of iterations r_{iter}

1: Normalize input HSI $\mathbf{H}$ according to (1);
2: Calculate the guided image $\mathbf{G}$ according to Algorithm 1;
3: Train the proposed guided AE network by the back-propagation method by combining (9) and (10);
4: Reconstruct the HSI $\mathbf{H}$ according to (2)–(5);
5: Extract the anomaly targets according to (11);

Output: Anomaly detection result $\mathbf{F}$

IV. Experimental Results and Analysis

This section evaluates the effectiveness and superiority of the proposed method by implementing different experiments on six real-world datasets. First, the experimental datasets and evaluation indicators are introduced. Second, the parameter setting of the comparison methods and the proposed method are analyzed. Then, the results of the proposed method and the results of the comparison methods on six datasets are compared and analyzed. Finally, other performances of the proposed method are further analyzed, including the performance of the guided module, noise robustness analysis, and small target detection ability. Each experiment was performed using MATLAB 2017b on a computer with an Intel Core i5-9400 CPU and 16-GB RAM.

A. Experimental Datasets

1) San Diego: This dataset was collected at the airport in San Diego, CA, USA, by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) [37]. This HSI is 400 × 400 in size. There are 224 bands with a 10-nm spectral resolution, and its band range is from 0.37 to 2.51 μm. After removing the low signal-to-noise ratio (SNR) and water absorption bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224), 189 bands were retained in the experiments. Two 100 × 100 size regions were cropped from the HSI and applied in the experiments. The false-color images of the two subimages and the corresponding ground-truth images are shown in Fig. 3(a) and (b), respectively. Aircrafts are treated as anomaly targets in each HSI. The two HSIs are denoted as San Diego I and San Diego II, respectively.

2) Pavia: This dataset was acquired in Pavia, Italy, with the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor [47]. This HSI has 102 bands with a spatial resolution of 1.3 m/pixel. In the experiments, a subimage of 100 × 100 pixels was cropped from this HSI, and all the bands were reserved. The cars in the HSI are considered anomaly targets. The false-color image of this HSI and the corresponding ground-truth image are shown in Fig. 3(c).

3) Texas Coast: This dataset was collected at the Texas Coast, USA, by the AVIRIS [47]. This HSI has 100 × 100 pixels, 207 bands, and a spatial resolution of 17.2 m/pixel. All the bands were used in each experiment. The false-color image of this HSI and the corresponding ground-truth image are shown in Fig. 3(d).

4) Salinas: This dataset was collected by the AVIRIS at the Salinas Valley, CA, USA [48]. A total of 512 × 217 pixels and 224 bands are included in this HSI, and its spatial resolution is 3.7 m/pixel. The low SNR and water absorption bands (108–112, 154–167, and 224) were removed in the experiments. An HSI, containing 200 × 150 pixels, was cropped for the experiments from the original HSI. The manufactured structures in this HSI are considered anomaly targets. The false-color image of the HSI and the corresponding ground-truth image are shown in Fig. 3(e).

5) MUUFL Gulfport: This HSI was collected at the University of Southern Mississippi Gulf Park Campus, Long Beach, MS, USA, by the hyperspectral compact airborne spectrographic imager (CASI–1500) [49]. The original HSI

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

size is 325 × 220 × 72. The first four and the last four noise bands were removed in the experiments. A 200 × 220 scene was cropped from the original HSI. Therefore, the size of the HSI used in the experiment is 200 × 220 × 64. In this HSI, four different colored cloth panels are considered the anomaly targets. The false-color image of the HSI and the corresponding ground-truth image are shown in Fig. 3(f).

B. Evaluation Indicator

The experiment compares the proposed method with the following state-of-the-art methods: GRX [12], LRX [13], CRD [19], LRASR [23], AE [31], RGAE [31], and Auto-AD [32] methods. The AE method is a traditional AE framework with a single hidden layer. The MSE is used as the loss function, and the reconstruction error is used as an anomaly detection result. The proposed guided AE-based detector is denoted as GAED.

The detection results of all the methods were exploited for qualitative evaluation. The three-dimensional receiver operating characteristic (3-D ROC) curve [50], the corresponding two-dimensional ROC (2-D ROC) curve, and the area under the 2-D ROC curve (AUC) were used as quantitative evaluation indicators. The 3-D ROC curve is a function of detection probability P_{D}, false alarm probability P_{F}, and threshold τ. The 3-D ROC curve can be decomposed into three 2-D ROC curves, which are $(P_{\mathrm{D}}, P_{\mathrm{F}})$, (P_{D}, τ), and (P_{F}, τ), respectively. The AUC values of these three 2-D ROC curves are denoted as $\mathrm{AUC}_{(\mathrm{D,F})}$, $\mathrm{AUC}_{(\mathrm{D},\tau)}$, and $\mathrm{AUC}_{(\mathrm{F},\tau)}$, respectively. Five other AUC values are generated from these three AUC values, denoted as $\mathrm{AUC}_{\mathrm{TD}}$, $\mathrm{AUC}_{\mathrm{BS}}$, $\mathrm{AUC}_{\mathrm{SNPR}}$, $\mathrm{AUC}_{\mathrm{TDBS}}$, and $\mathrm{AUC}_{\mathrm{ODP}}$, respectively. The detailed calculation method is

$$\begin{cases} \mathrm{AUC}_{\mathrm{TD}} &= \mathrm{AUC}_{(\mathrm{D,F})} + \mathrm{AUC}_{(\mathrm{D},\tau)} \\ \mathrm{AUC}_{\mathrm{BS}} &= \mathrm{AUC}_{(\mathrm{D,F})} - \mathrm{AUC}_{(\mathrm{F},\tau)} \\ \mathrm{AUC}_{\mathrm{SNPR}} &= \dfrac{\mathrm{AUC}_{(\mathrm{D},\tau)}}{\mathrm{AUC}_{(\mathrm{F},\tau)}} \\ \mathrm{AUC}_{\mathrm{TDBS}} &= \mathrm{AUC}_{(\mathrm{D},\tau)} - \mathrm{AUC}_{(\mathrm{F},\tau)} \\ \mathrm{AUC}_{\mathrm{ODP}} &= \mathrm{AUC}_{(\mathrm{D,F})} + \mathrm{AUC}_{(\mathrm{D},\tau)} - \mathrm{AUC}_{(\mathrm{F},\tau)}. \end{cases} \tag{12}$$

$\mathrm{AUC}_{(\mathrm{D,F})}$, $\mathrm{AUC}_{\mathrm{TDBS}}$, and $\mathrm{AUC}_{\mathrm{ODP}}$ are used to evaluate the overall performance of a detector. $\mathrm{AUC}_{(\mathrm{D},\tau)}$ and $\mathrm{AUC}_{\mathrm{TD}}$ are used to evaluate the target detection probability of a detector. $\mathrm{AUC}_{(\mathrm{F},\tau)}$, $\mathrm{AUC}_{\mathrm{BS}}$, and $\mathrm{AUC}_{\mathrm{SNPR}}$ are used to evaluate the background suppression ability of a detector. The smaller the value of $\mathrm{AUC}_{(\mathrm{F},\tau)}$, the better. The larger the other AUC values, the better.

C. Parameter Setting

The uncertain parameters of the state-of-the-art methods in the experiment are set according to the suggestions of the original authors or $\mathrm{AUC}_{(\mathrm{D,F})}$ value. The GRX method has no additional uncertain parameters. In the LRX method, the double-window size is set to (13, 17), (13, 15), (11, 13), (13, 17), (15, 17), and (13, 31) on the six datasets, respectively. In the CRD method, the double-window size is set to (11, 13), (11, 15), (7, 9), (5, 7), (9, 11), and (19, 21) on the six datasets, respectively. The regularization parameter is set to 10^{-6} on all the datasets. In the dictionary construction of the LRASR method, the number of categories and selected pixels is set to 20 and 15 for all the datasets, respectively. The two regularization parameters β and λ are set to (0.1, 0.1), (0.01, 0.01), (0.01, 0.01), (0.01, 0.01), (0.1, 0.1), and (0.1, 0.1), respectively. In the AE method, for all the datasets, the weights and biases are randomly initialized, the number of hidden layer nodes is set to 25, the learning rate is 0.1, and the maximum number of iterations is 300. In the RGAE methods, for all the datasets, the weights and biases are randomly initialized, the tradeoff parameter is 0.01, the dimension of the hidden layer is 100, the number of superpixels is 150, the maximum number of iterations is 300, the learning rate is 0.01, and the maximum number of iterations is 1200. Since the Auto-AD method is related to autonomous hyperspectral anomaly detection, it sets no additional parameters.

All the weights and biases in the proposed GAED method were randomly initialized. The dimensions of the second to sixth layers were 100, 50, 25, 50, and 100, respectively. We used sigmoid as the activation function and MSE as the loss function. The hyperparameters in the proposed GAED method were window size c, learning rate l_{a}, penalty coefficient β, and number of iterations r_{iter}. In analyzing the influence of various parameters on the performance of the proposed GAED method, $\mathrm{AUC}_{(\mathrm{D,F})}$ value was used as an evaluation metric. To analyze a particular hyperparameter, all other hyperparameters were held constant. The default values were set as $c = 7$, $l_{\mathrm{a}} = 0.1$, $\beta = 1$, and $r_{\mathrm{iter}} = 300$.

1) Window Size (c): The window size was found to be affecting the guided image quality. For all the datasets, the parameters l_{a}, β, and r_{iter} were set to 0.1, 1, and 300, respectively. The value of c was varied from 3 to 13, and its effect on the performance of the proposed method is shown in Fig. 4(a). According to Fig. 4(a), the proposed method remained relatively stable for various values of c. Finally, we set c to 7 for all the datasets.

2) Learning Rate (l_a): For all the datasets, the parameters c, β, and r_{iter} were set to 7, 1, and 300, respectively. The value of r_{iter} was varied from 0.1 to 0.9. Fig. 4(b) shows the effect of variable l_{a} on the performance of the proposed method. We set l_{a} to 0.4, 0.1, 0.4, 0.1, 0.4, and 0.4 for each dataset, respectively.

3) Penalty Coefficient (β): To determine β for each dataset, other parameters c, l_{a}, and r_{iter} were set to 7, 0.1, and 300, respectively. As indicated in Fig. 4(c), the performance of the proposed method was evaluated for β at $10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1$, and 10, respectively. Finally, β was set to 1 for the first dataset and 10 for the other datasets.

4) Number of Iterations (r_{iter}): For all the datasets, parameters c, l_{a}, and β were set to 7, 0.1, and 1, respectively. To determine an optimal value of r_{iter}, we used a range of values varying from 100 to 1000. According to Fig. 4(d), r_{iter} was set to 300, 100, 500, 100, 300, and 500, respectively, for each dataset.

5) Dimension of the Middle-Hidden Layer: By keeping all other parameters constant, we came up with an optimal dimension for the middle hidden layer of the network.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

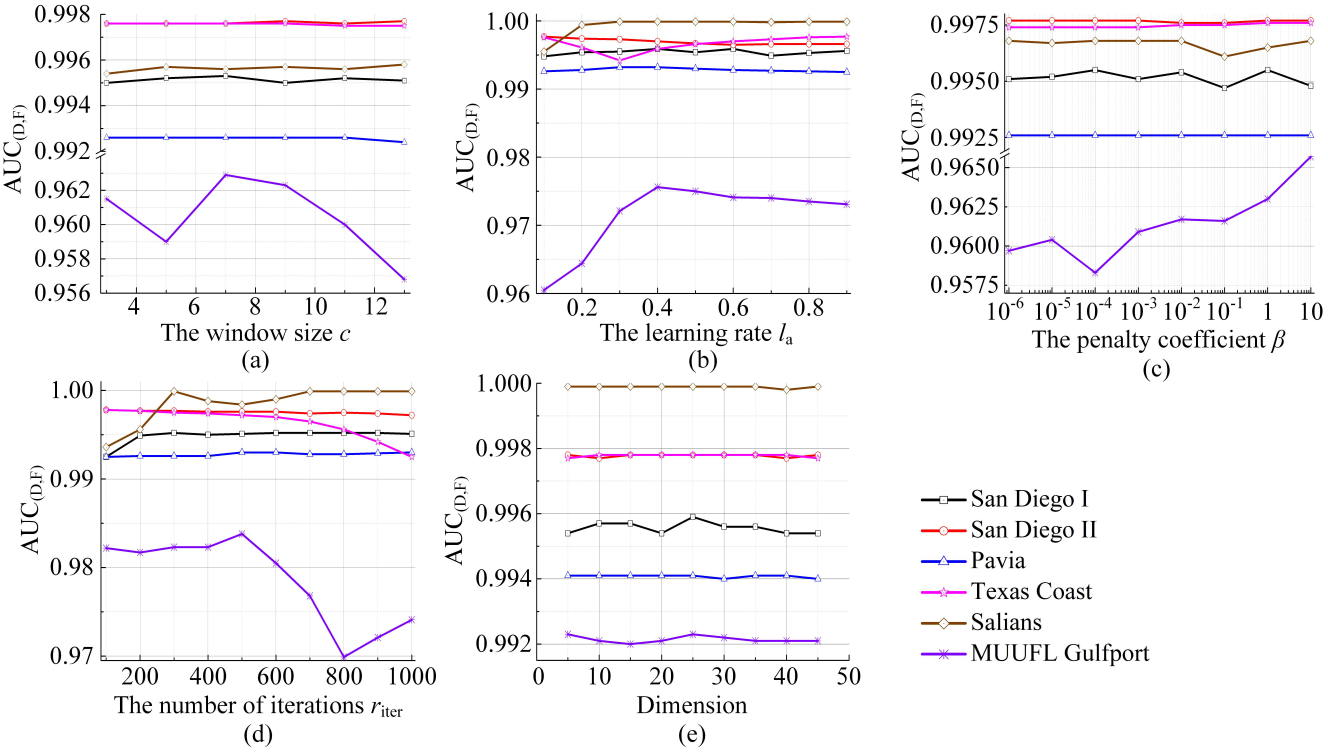

Fig. 4. Effect of different parameters on the performance of the proposed method. (a) Window size c. (b) Learning rate l_a. (c) Penalty coefficient β. (d) Number of iterations r_{iter}. (e) Dimension of the AE middle hidden layer.

The tested dimensions were 5, 10, 15, 20, 25, 30, 35, 40, and 45, respectively. A change in network hidden dimensions did not significantly affect the detection performance as shown in Fig. 4(e), The experiments on all the datasets was conducted with a network having 25-D middle hidden layer.

D. Experimental Results

This section analyzes the detection performance of the proposed GAED method by comparing it with seven other state-of-the-art detection methods. The detection results of all the methods on different datasets are shown in Fig. 5. On the first dataset, the GRX method does not separate anomaly targets from the background. Although the targets are detected on other datasets, the background part is also clearer. The LRX method separates the targets from the background on the Pavia dataset but does not separate the targets from the background on other datasets. There are some false alarms in the detection results of the LRX method. Both the CRD and LRASR methods detect anomaly targets on all the datasets, but the background is also relatively clear in their detection results. In their detection results, the background suppression effect is not satisfactory. Both the AE and RGAE methods successfully detect the anomaly targets from the background. The background suppression effect of their detection results is satisfactory. The detection results of the AE and RGAE methods are significantly close to the detection results of the proposed GAED method. In the first and last datasets, although the Auto-AD method detects anomaly targets, it does not successfully separate the background and anomaly targets. On other datasets, the Auto-AD method successfully detects the anomaly targets, and the background suppression effect of the detection results is also satisfactory. The proposed GAED method successfully detects the anomaly targets and separates the anomaly targets from the background on all the datasets. The background suppression effect of the detection results of the proposed GAED method is satisfactory. Overall, from the visual detection results, the proposed GAED method obtains excellent detection results across all the datasets.

The 3-D ROC curves and the corresponding AUC values of the detection results of all the methods on the San Diego I dataset are shown in Fig. 6 and Table I, respectively. From the 3-D ROC curves and the corresponding 2-D ROC curves in Fig. 6, on San Diego I, the ROC curve of the proposed GAED method is higher than most other ROC curves. In addition, $\text{AUC}_{(\text{D,F})}$, AUC_{BS}, AUC_{SNPR}, AUC_{TDBS}, and AUC_{ODP} of the proposed GAED method in Table I are larger than those of the comparison methods, which are 0.9959, 0.9887, 28.2004, 0.1933, and 1.1892, respectively. $\text{AUC}_{(\text{D},\tau)} = 0.3012$ and $\text{AUC}_{\text{TD}} = 1.2788$ of the CRD method are larger than the other corresponding AUC values. $\text{AUC}_{(\text{F},\tau)} = 0.0069$ of the Auto-AD method is the best among the proposed GAED method and all the comparison methods. Although $\text{AUC}_{(\text{D},\tau)}$, AUC_{TD}, and $\text{AUC}_{(\text{F},\tau)}$ values of the proposed GAED method are not the best, they are also very close to the optimal value.

The 3-D ROC curves and the corresponding AUC values of the detection results of the comparison methods and the proposed method on San Diego II are shown in Fig. 7

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

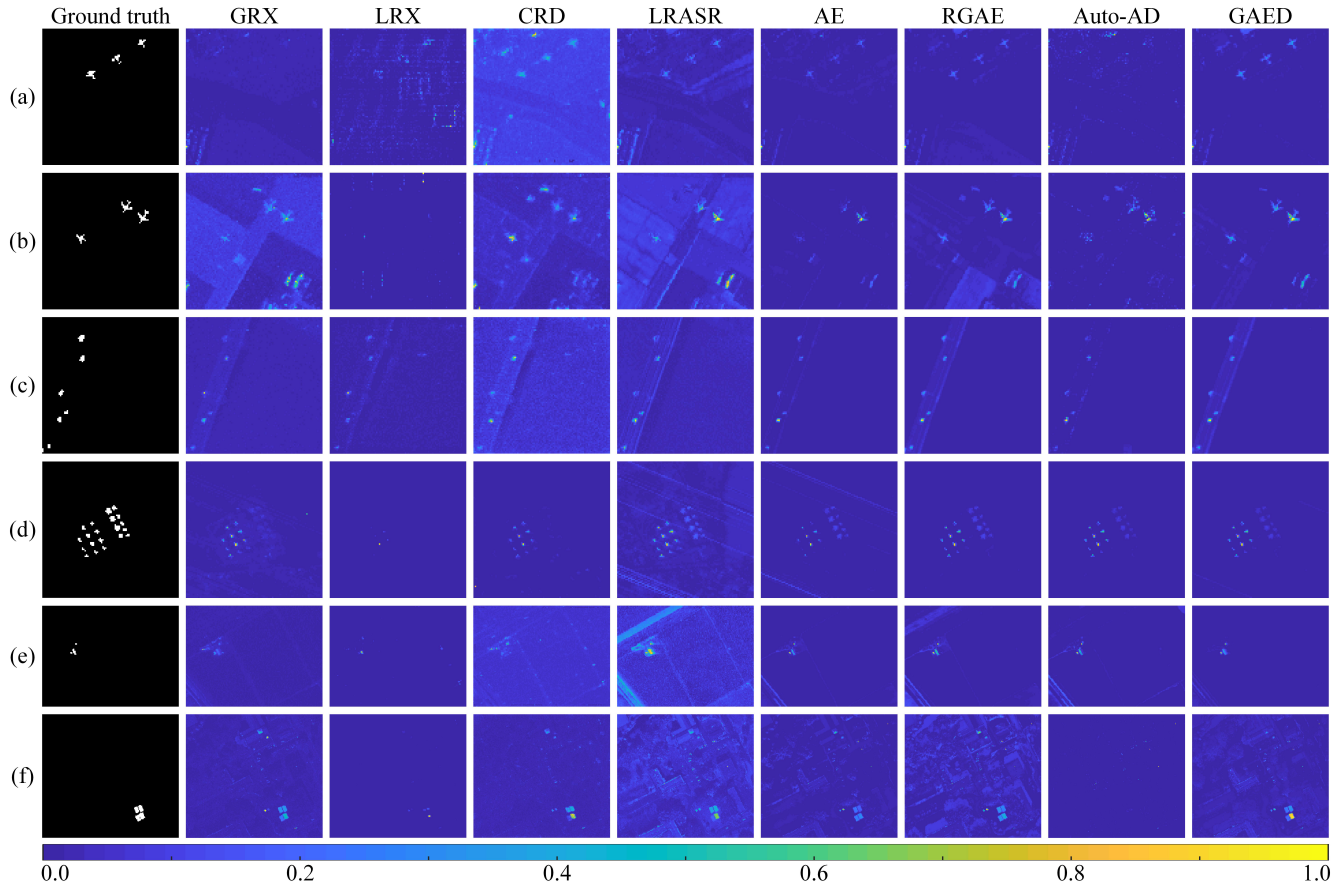

Fig. 5. Detection results of different methods on each dataset. (a) San Diego I. (b) San Diego II. (c) Pavia. (d) Texas Coast. (e) Salinas. (f) MUUFL Gulfport.

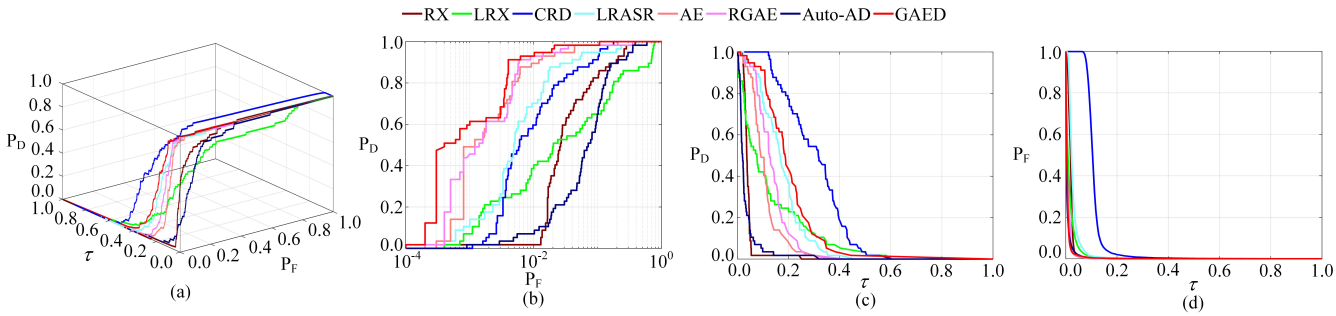

Fig. 6. Performance comparison of the 3-D ROC curves of different methods on San Diego I. (a) Three-dimensional ROC curves. (b) Corresponding 2-D ROC curves (P_D, P_F). (c) Corresponding 2-D ROC curves (P_D, τ). (d) Corresponding 2-D ROC curves (P_F, τ).

TABLE I

AUC PERFORMANCE COMPARISON OF DIFFERENT METHODS ON SAN DIEGO I

Method	$AUC_{(D,F)}$ ↑	$AUC_{(D,\tau)}$ ↑	$AUC_{(F,\tau)}$ ↓	AUC_{TD} ↑	AUC_{BS} ↑	AUC_{SNPR} ↑	AUC_{TDBS} ↑	AUC_{ODP} ↑
RX	0.9398	0.0411	0.0193	0.9808	0.9205	2.1313	0.0218	0.9616
LRX	0.8585	0.1243	0.0157	0.9828	0.8428	7.9089	0.1086	0.9671
CRD	0.9776	**0.3012**	0.1101	**1.2788**	0.8675	2.7359	0.1911	1.1687
LRASR	0.9764	0.1712	0.0283	1.1477	0.9481	6.0503	0.1429	1.1194
AE	0.9922	0.1055	0.0074	1.0977	0.9847	14.1820	0.0981	1.0902
RGAE	0.9904	0.1337	0.0083	1.1241	0.9821	16.0775	0.1254	1.1158
Auto-AD	0.9019	0.0324	**0.0069**	0.9342	0.8949	4.6773	0.0255	0.9273
GAED	**0.9959**	0.2005	0.0071	1.1963	**0.9887**	**28.2004**	**0.1933**	**1.1892**

and Table II. From the 3-D ROC curves and the corresponding 2-D ROC curves in Fig. 7, the comprehensive performance of the proposed GAED method is excellent. As shown in Table II, $AUC_{(D,F)} = 0.9978$, $AUC_{BS} = 0.9894$, $AUC_{SNPR} = 38.0194$, $AUC_{TDBS} = 0.3100$, and $AUC_{ODP} = 1.3078$ of the proposed GAED method are better than those of other

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

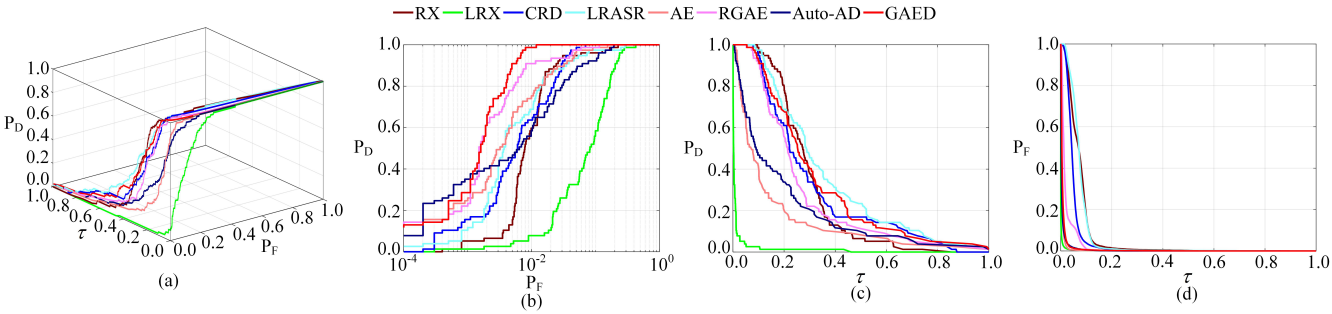

Fig. 7. Performance comparison of the 3-D ROC curves of different methods on San Diego II. (a) Three-dimensional ROC curves. (b) Corresponding 2-D ROC curves (P_D, P_F). (c) Corresponding 2-D ROC curves (P_D, τ). (d) Corresponding 2-D ROC curves (P_F, τ).

TABLE II
AUC PERFORMANCE COMPARISON OF DIFFERENT METHODS ON SAN DIEGO II

Method	$AUC_{(D,F)}$ ↑	$AUC_{(D,\tau)}$ ↑	$AUC_{(F,\tau)}$ ↓	AUC_{TD} ↑	AUC_{BS} ↑	AUC_{SNPR} ↑	AUC_{TDBS} ↑	AUC_{ODP} ↑
RX	0.9833	0.2889	0.0714	1.2722	0.9120	4.0480	0.2175	1.2008
LRX	0.9010	0.0129	**0.0017**	0.9139	0.8993	7.5740	0.0112	0.9122
CRD	0.9885	0.2982	0.0483	1.2867	0.9402	6.1794	0.2500	1.2384
LRASR	0.9835	**0.3537**	0.0732	**1.3372**	0.9103	4.8323	0.2805	1.2640
AE	0.9892	0.1481	0.0050	1.1373	0.9841	29.4227	0.1431	1.1323
RGAE	0.9937	0.2635	0.0206	1.2571	0.9731	12.8189	0.2429	1.2366
Auto-AD	0.9782	0.1894	0.0055	1.1676	0.9727	34.6041	0.1840	1.1621
GAED	**0.9978**	0.3184	0.0084	1.3162	**0.9894**	**38.0194**	**0.3100**	**1.3078**

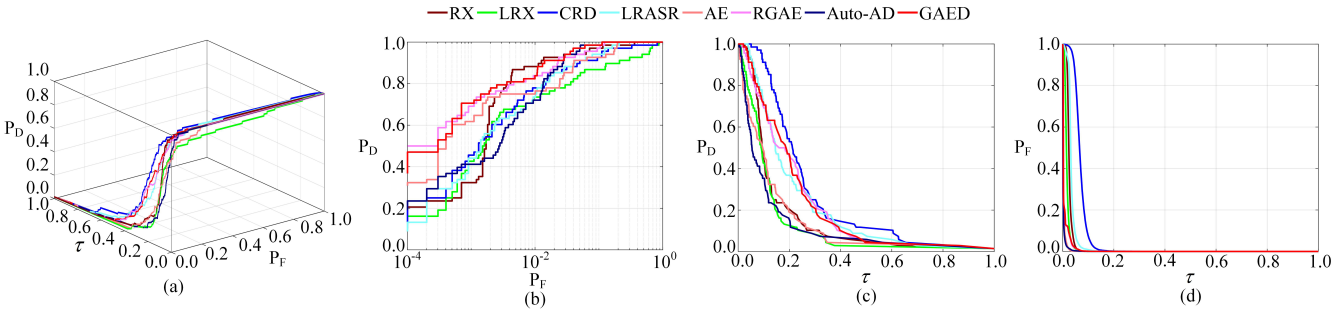

Fig. 8. Performance comparison of the 3-D ROC curves of different methods on Pavia. (a) Three-dimensional ROC curves. (b) Corresponding 2-D ROC curves (P_D, P_F). (c) Corresponding 2-D ROC curves (P_D, τ). (d) Corresponding 2-D ROC curves (P_F, τ).

comparison methods. In this experiment, $AUC_{(D,\tau)} = 0.3537$ and $AUC_{TD} = 1.3372$ of the LRASR method are better than those of other methods, and $AUC_{(F,\tau)} = 0.0017$ of the LRX method is better than that of other methods. In addition, $AUC_{(D,\tau)}$, $AUC_{(F,\tau)}$, and AUC_{TD} of the proposed GAED method are also close to the corresponding best AUC values among all the methods in the experiment.

The 3-D ROC curves and the corresponding AUC values of the detection results of the comparison methods and the proposed method on Pavia are shown in Fig. 8 and Table III. As shown in Table III, $AUC_{(D,F)}$ and AUC_{ODP} values of the proposed GAED method are better than those of other comparative methods, 0.9941 and 1.2030, respectively. The CRD method obtained the best $AUC_{(D,\tau)}$ and AUC_{TD} values, which were 0.2642 and 1.2359, respectively. The RGAE method obtained the best AUC_{SNPR} and AUC_{TDBS} values in all the methods in this experiment, which were 45.3351 and 0.2098, respectively. The Auto-AD method obtained the best $AUC_{(F,\tau)}$ and AUC_{BS} values in all the methods in this experiment, which were 0.0026 and 0.9822, respectively. In addition, it can be seen from Table III that although $AUC_{(D,\tau)}$, $AUC_{(F,\tau)}$, AUC_{TD}, the AUC_{BS}, AUC_{SNPR}, and AUC_{TDBS} values of the proposed GAED method are not optimal, they are also close to the optimal AUC values.

The 3-D ROC curves and the corresponding AUC values of the detection results of the comparison methods and the proposed method on Texas Coast are shown in Fig. 9 and Table IV. From the 3-D ROC curves and the corresponding 2-D ROC curves in Fig. 9, the ROC curves of the proposed GAED method are relatively close to the ROC curves of other comparison methods. In Table IV, the best AUC values are $AUC_{(F,\tau)} = 0.0002$ of the LRX method, $AUC_{(D,\tau)} = 0.1847$, $AUC_{TD} = 1.1609$, $AUC_{TDBS} = 0.1582$, and $AUC_{ODP} = 1.1345$ of the LRASR method, $AUC_{(D,F)} = 0.9994$ of the RGAE method, and $AUC_{BS} = 0.9973$ and $AUC_{SNPR} = 48.9754$ of the Auto-AD method. In Table IV, although all the AUC values of the proposed GAED method are not optimal, they are slightly below the optimal AUC values.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

TABLE III

AUC PERFORMANCE COMPARISON OF DIFFERENT METHODS ON PAVIA

Method	$AUC_{(D,F)}$ ↑	$AUC_{(D,\tau)}$ ↑	$AUC_{(F,\tau)}$ ↓	AUC_{TD} ↑	AUC_{BS} ↑	AUC_{SNPR} ↑	AUC_{TDBS} ↑	AUC_{ODP} ↑
RX	0.9887	0.1533	0.0259	1.1419	0.9628	5.9284	0.1274	1.1161
LRX	0.9348	0.1254	0.0165	1.0603	0.9184	7.6194	0.1090	1.0438
CRD	0.9717	**0.2642**	0.0686	**1.2359**	0.9032	3.8539	0.1956	1.1674
LRASR	0.9840	0.2144	0.0339	1.1984	0.9501	6.3220	0.1805	1.1645
AE	0.9815	0.1375	0.0035	1.1189	0.9779	38.8573	0.1340	1.1154
RGAE	0.9914	0.2145	0.0047	1.2059	0.9867	**45.3351**	**0.2098**	1.2012
Auto-AD	0.9908	0.1226	**0.0026**	1.1134	**0.9882**	47.5666	0.1200	1.1108
GAED	**0.9941**	0.2160	0.0070	1.2101	0.9870	30.7169	0.2090	**1.2030**

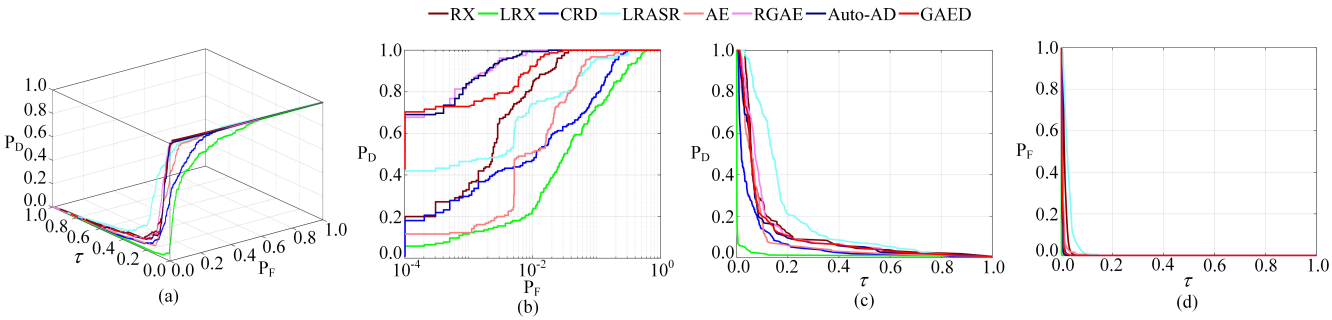

Fig. 9. Performance comparison of the 3-D ROC curves of different methods on Texas Coast. (a) Three-dimensional ROC curves. (b) Corresponding 2-D ROC curves (P_D, P_F). (c) Corresponding 2-D ROC curves (P_D, τ). (d) Corresponding 2-D ROC curves (P_F, τ).

TABLE IV

AUC PERFORMANCE COMPARISON OF DIFFERENT METHODS ON TEXAS COAST

Method	$AUC_{(D,F)}$ ↑	$AUC_{(D,\tau)}$ ↑	$AUC_{(F,\tau)}$ ↓	AUC_{TD} ↑	AUC_{BS} ↑	AUC_{SNPR} ↑	AUC_{TDBS} ↑	AUC_{ODP} ↑
RX	0.9946	0.1179	0.0135	1.1124	0.9811	8.7151	0.1043	1.0989
LRX	0.9067	0.0133	**0.0002**	0.9201	0.9066	78.5001	0.0132	0.9199
CRD	0.9504	0.0625	0.0058	1.0129	0.9446	10.8265	0.0567	1.0071
LRASR	0.9763	**0.1847**	0.0265	**1.1609**	0.9498	6.9791	**0.1582**	**1.1345**
AE	0.9763	0.0830	0.0059	1.0593	0.9705	14.1860	0.0771	1.0535
RGAE	**0.9994**	0.1168	0.0027	1.1161	0.9966	42.8463	0.1141	1.1134
Auto-AD	0.9993	0.0996	0.0020	1.0989	**0.9973**	**48.9754**	0.0976	1.0969
GAED	0.9979	0.1035	0.0031	1.1013	0.9947	33.8752	0.1004	1.0982

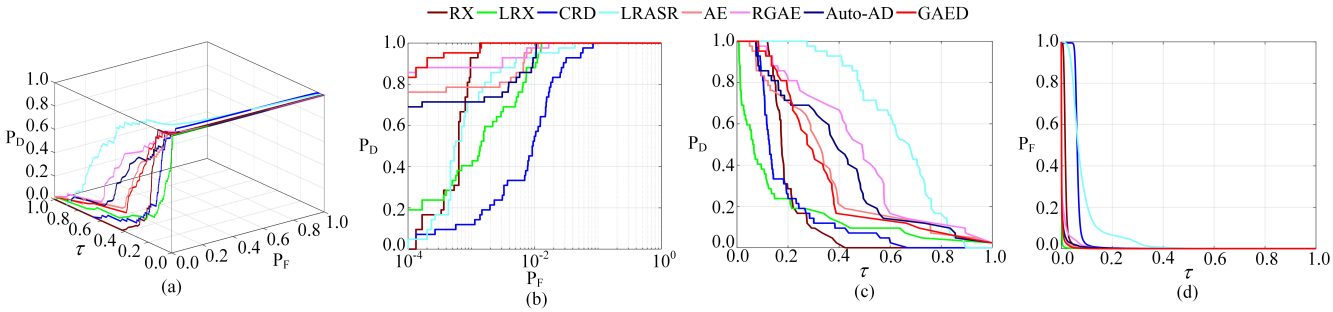

Fig. 10. Performance comparison of the 3-D ROC curves of different methods on Salinas. (a) Three-dimensional ROC curves. (b) Corresponding 2-D ROC curves (P_D, P_F). (c) Corresponding 2-D ROC curves (P_D, τ). (d) Corresponding 2-D ROC curves (P_F, τ).

The 3-D ROC curves and the corresponding AUC values of the detection results of the comparison methods and the proposed method on Salinas are shown in Fig. 10 and Table V. Table V shows that $AUC_{(D,F)}$, AUC_{BS}, and AUC_{SNPR} of the proposed GAED method are superior to those of other comparison methods, which are 0.9999, 0.9971, and 120.4167, respectively. $AUC_{(D,\tau)}$, AUC_{TD}, AUC_{TDBS}, and AUC_{ODP} of the LRASR method are superior to those of other comparison methods and the proposed GAED method, which are 0.6332, 1.6305, 0.5472, and 1.5445, respectively. $AUC_{(F,\tau)} = 0.0019$ of the LRX method is below that of the other methods in Table V. On the Salinas dataset, the performance of the proposed GAED method is slightly weaker than that of the LRASR method; however, the performance of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

TABLE V
AUC PERFORMANCE COMPARISON OF DIFFERENT METHODS ON SALINAS

Method	$AUC_{(D,F)}$ ↑	$AUC_{(D,\tau)}$ ↑	$AUC_{(F,\tau)}$ ↓	AUC_{TD} ↑	AUC_{BS} ↑	AUC_{SNPR} ↑	AUC_{TDBS} ↑	AUC_{ODP} ↑
RX	0.9994	0.1958	0.0175	1.1951	0.9818	11.1745	0.1783	1.1776
LRX	0.9968	0.1599	**0.0019**	1.1567	0.9949	83.3369	0.1580	1.1548
CRD	0.9861	0.1863	0.0650	1.1724	0.9211	2.8653	0.1213	1.1074
LRASR	0.9973	**0.6332**	0.0860	**1.6305**	0.9112	7.3597	**0.5472**	**1.5445**
AE	0.9986	0.3454	0.0041	1.3439	0.9944	83.5284	0.3412	1.3398
RGAE	0.9991	0.4658	0.0075	1.4649	0.9916	61.9026	0.4582	1.4573
Auto-AD	0.9980	0.3950	0.0045	1.3931	0.9936	88.6357	0.3906	1.3886
GAED	**0.9999**	0.3381	0.0028	1.3380	**0.9971**	**120.4167**	0.3353	1.3352

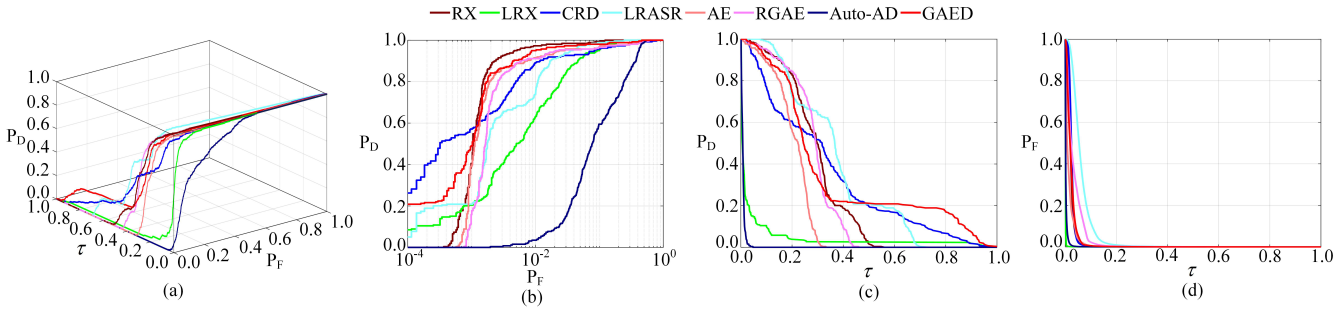

Fig. 11. Performance comparison of the 3-D ROC curves of different methods on MUUFL Gulfport. (a) Three-dimensional ROC curves. (b) Corresponding 2-D ROC curves (P_D, P_F). (c) Corresponding 2-D ROC curves (P_D, τ). (d) Corresponding 2-D ROC curves (P_F, τ).

TABLE VI
AUC PERFORMANCE COMPARISON OF DIFFERENT METHODS ON MUUFL GULFPORT

Method	$AUC_{(D,F)}$ ↑	$AUC_{(D,\tau)}$ ↑	$AUC_{(F,\tau)}$ ↓	AUC_{TD} ↑	AUC_{BS} ↑	AUC_{SNPR} ↑	AUC_{TDBS} ↑	AUC_{ODP} ↑
RX	**0.9970**	0.2920	0.0224	1.2890	0.9746	13.0220	0.2695	1.2666
LRX	0.9796	0.0453	**0.0011**	1.0249	**0.9786**	**41.4939**	0.0442	1.0238
CRD	0.9810	0.3232	0.0258	1.3042	0.9552	12.5224	0.2974	1.2784
LRASR	0.9911	**0.3646**	0.0617	**1.3557**	0.9294	5.9105	0.3029	1.2940
AE	0.9863	0.2028	0.0146	1.1891	0.9717	13.9343	0.1883	1.1746
RGAE	0.9848	0.2814	0.0338	1.2662	0.9511	8.3362	0.2477	1.2325
Auto-AD	0.8630	0.0082	0.0027	0.8712	0.8604	3.0729	0.0055	0.8685
GAED	0.9923	0.3530	0.0222	1.3453	0.9700	15.8799	**0.3308**	**1.3231**

the proposed GAED method approaches that of the LRASR method.

The 3-D ROC curves and the corresponding AUC values of the detection results of the comparison methods and the proposed method on MUUFL Gulfport are shown in Fig. 11 and Table VI. In Fig. 11, since the ROC curves of the proposed method are mixed with those of some comparison methods, it is difficult to distinguish which method has the best performance. Hence, we need to compare the performance of all the methods combined with the corresponding AUC values in Table VI. In Table VI, the best AUC values for each method are $AUC_{(D,F)}$ value of the RX method, $AUC_{(F,\tau)}$, AUC_{BS}, and AUC_{SNPR} values of the LRX method, $AUC_{(D,\tau)}$ and AUC_{TD} values of the LRASR method, and AUC_{TDBS} and AUC_{ODP} values of the proposed GAED method, respectively. These optimal AUC values are 0.9970, 0.0011, 0.9786, 41.4939, 0.3646, 1.3557, 0.3308, and 1.3231, respectively. Although the AUC values of the proposed GAED method are not all the best, except for AUCTDBS and AUCODP, other AUC values are close to the best AUC values. On all the datasets in the experiment, the performance of the proposed GAED method is not all optimal, but it achieves satisfactory results. Therefore, the proposed GAED method is an effective and excellent anomaly detection method.

In addition, we present the efficiency of the proposed method and comparative methods on six datasets. All the methods were performed using MATLAB 2017b on a computer with an Intel Core i5-9400 CPU and 16-GB RAM. The efficiency of the different anomaly detection methods on six datasets is shown in Table VII. It is worth noting that this time includes the training time and test time. The GRX method has the shortest running time on all the datasets. The running time of the proposed GAED method on all the datasets is between the longest time and the shortest time.

E. Performance Analysis of the Guided Module

To analyze the performance of the guided module, we remove the guided module in the proposed GAED method and only use the proposed AE network as the anomaly detection method. The visual detection results and the AUC values calculated from the 3-D ROC curves were used as evaluation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

TABLE VII

EFFICIENCY OF THE DIFFERENT ANOMALY DETECTION METHODS ON SIX DATASETS

Method	Time (s)					
	San Diego I	San Diego II	Pavia	Texas Coast	Salinas	MUUFL Gulfport
RX	1.7	1.6	1.0	1.0	1.7	2.0
LRX	43.9	35.3	13.1	49.9	125.1	114.5
CRD	36.5	48.9	13.0	36.3	113.8	76.7
LRASR	25.9	23.6	19.4	18.3	109.5	266.6
AE	24.2	23.2	12.4	24.3	57.4	38.9
RGAE	83.6	83.7	31.3	89.3	281.5	198.1
Auto-AD	467.8	355.2	301.3	620.3	701.8	363.8
GAED	45.7	22.1	58.1	23.3	148.6	227.3

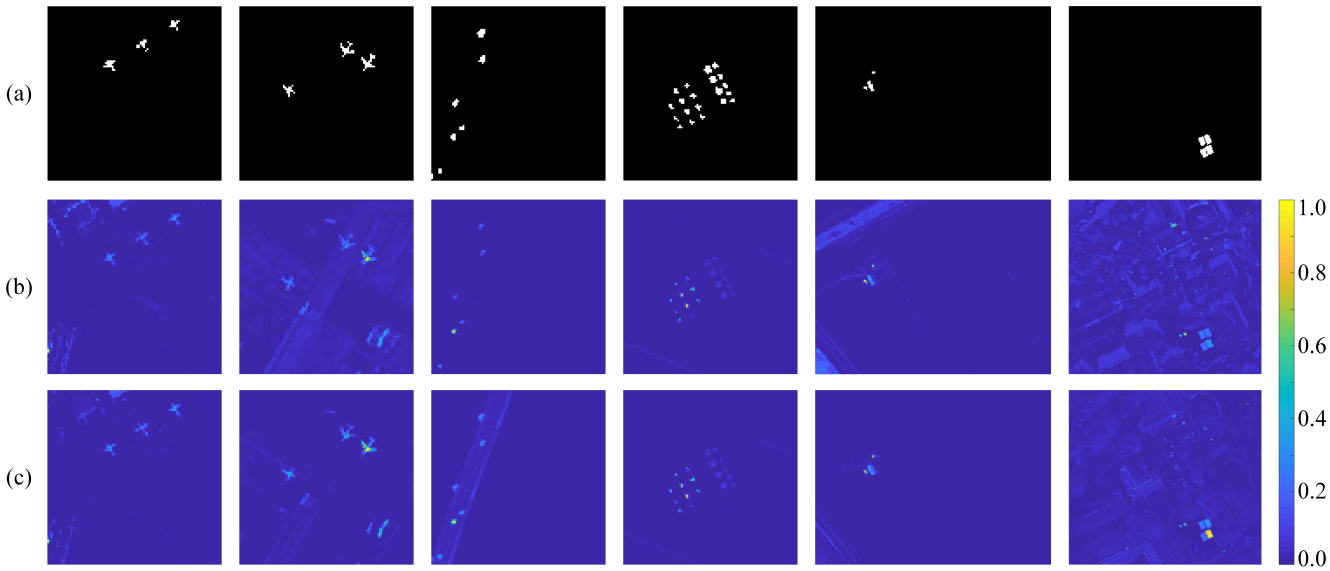

Fig. 12. Detection results of the proposed GAED method and the proposed method without the guided module on six datasets. (a) Ground truth. (b) Detection results of the proposed method without the guided module on six datasets. (c) Detection results of the proposed method with the guided module on six datasets.

TABLE VIII

AUC PERFORMANCE COMPARISON OF THE PROPOSED GAED METHOD AND THE PROPOSED METHOD WITHOUT THE GUIDED MODULE ON SIX DATASETS

Dataset	Guided Module	$AUC_{(D,F)}$ ↑	$AUC_{(D,\tau)}$ ↑	$AUC_{(F,\tau)}$ ↓	AUC_{TD} ↑	AUC_{BS} ↑	AUC_{SNPR} ↑	AUC_{TDBS} ↑	AUC_{ODP} ↑
San Diego I	✗	0.9923	0.1754	0.0085	1.1677	0.9838	20.5610	0.1669	1.1591
	✓	**0.9959**	**0.2005**	**0.0071**	**1.1963**	**0.9887**	**28.2004**	**0.1993**	**1.1892**
San Diego II	✗	0.9926	0.2520	0.0209	1.2445	0.9716	12.0386	0.2311	1.2236
	✓	**0.9978**	**0.3184**	**0.0084**	**1.3162**	**0.9894**	**38.0194**	**0.3100**	**1.3078**
Pavia	✗	0.9579	0.1233	**0.0014**	1.0812	0.9565	**85.4925**	0.1219	1.0798
	✓	**0.9941**	**0.2160**	0.0070	**1.2101**	**0.9870**	30.7169	**0.2090**	**1.2030**
Texas Coast	✗	0.9966	0.0946	**0.0026**	1.0911	0.9940	**36.2854**	0.0920	1.0885
	✓	**0.9978**	**0.1035**	0.0031	**1.1013**	**0.9947**	33.8752	**0.1004**	**1.0982**
Salinas	✗	0.9931	**0.3474**	0.0085	**1.3405**	0.9846	40.9953	**0.3390**	1.3321
	✓	**0.9999**	0.3381	**0.0028**	1.3380	**0.9971**	**120.4167**	0.3353	**1.3352**
MUUFL Gulfport	✗	0.9825	0.2167	0.0278	1.1992	0.9547	7.8061	0.1890	1.1714
	✓	**0.9923**	**0.3530**	**0.0222**	**1.3453**	**0.9700**	**15.8799**	**0.3308**	**1.3231**

indicators in the experiments. The detection results of the proposed GAED method and the proposed method without the guided module on six datasets are shown in Fig. 12. The corresponding AUC values are shown in Table VIII.

From the detection results in Fig. 12, the detection results of the two methods are similar. Both the methods successfully detect anomaly targets from the background. Among them, on the first two and the last two datasets, the background of the detection results of the proposed method without the guided module is slightly clearer than those of the detection results of the proposed GAED method. On the Pavia dataset, the background suppression effect of the detection results of the proposed method without the guided module is superior to that of the proposed GAED method. On the Texas Coast, the detection results of the two methods are similar. In Table VIII, on the first two and last datasets, all the AUC values of the proposed GAED method are larger than those without the guided module. On the Pavia and Texas Coast datasets, $AUC_{(F,\tau)}$ and AUC_{SNPR} of the proposed method without the guided module are better than the corresponding AUC

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

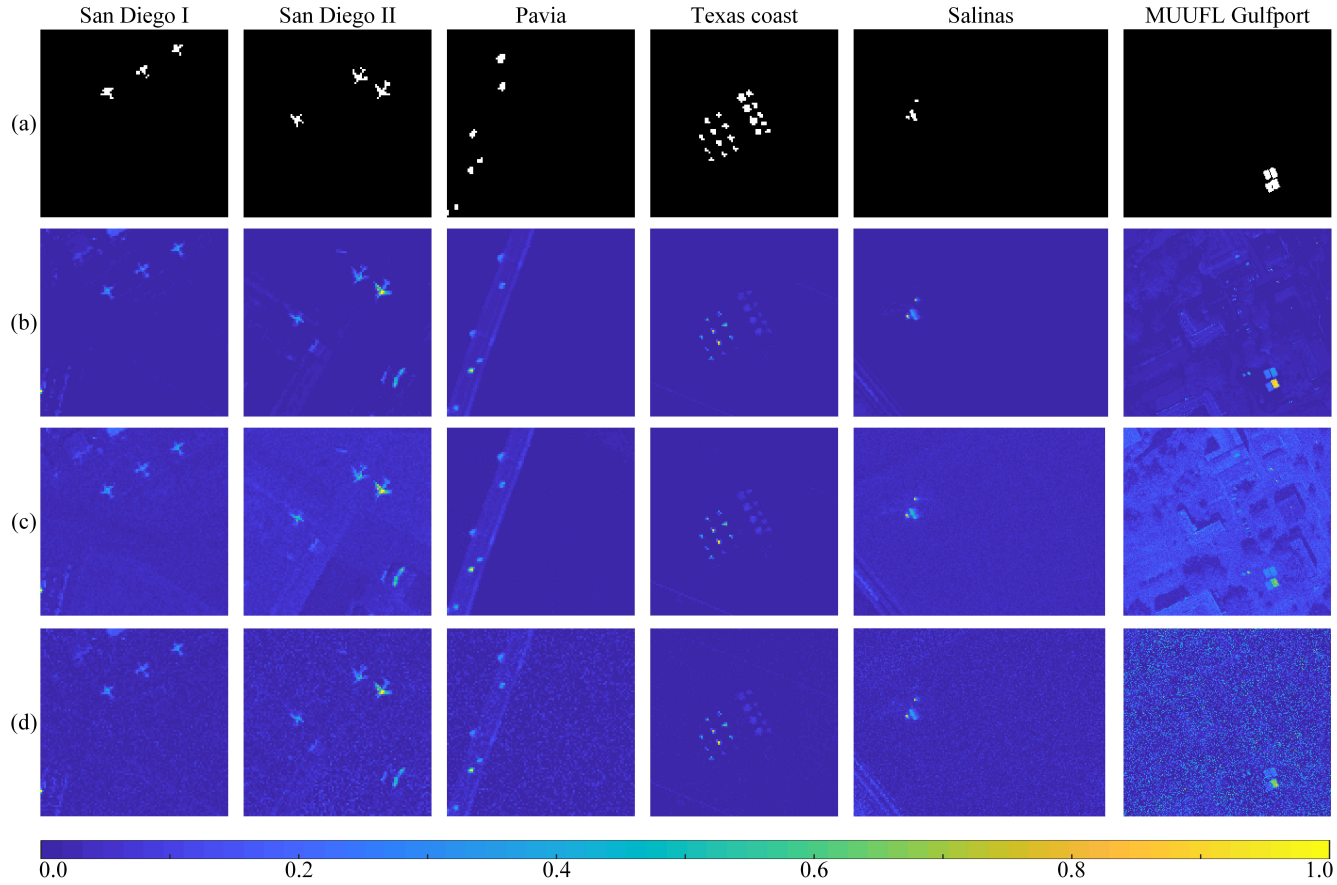

Fig. 13. Detection results from the proposed method on the original HSI and the noise-added HSI. (a) Ground-truth images. (b) Detection result on the original HSI. (c) Detection result on HSI after adding Gaussian noise. (d) Detection result on HSI after adding salt and pepper noise.

values of the proposed GAED method, and the other AUC values of the proposed method without the guided module are below those of the proposed GAED method. On the Salinas dataset, $\mathrm{AUC}_{(\mathrm{D},\tau)} = 0.3474$, $\mathrm{AUC}_{\mathrm{TD}} = 1.3405$, and $\mathrm{AUC}_{\mathrm{TDBS}} = 0.3390$ of the proposed method without the guided module are superior to the corresponding AUC values of the proposed GAED method; however, the other AUC values of the proposed GAED method are better than those of the proposed method without the guided module. Although some AUC values of the proposed GAED method are not optimal, they are also close to the optimal AUC values. The analysis of the experimental results showed that the guided module strengthens the representation ability of the proposed AE network for background features. Thus, the reconstructed background is closer to the background of the original HSI. Finally, through the reconstruction error, the background in the detection results is suppressed better. Therefore, the guided module improves the detection performance of the proposed method.

F. Noise Robustness Analysis

In this experiment, we added noise to each band of the hyperspectral dataset to construct an input HSI. The added noises were the Gaussian noise with mean 0 and variance 0.01, and salt and pepper noise with density 0.01. $\mathrm{AUC}_{(\mathrm{D,F})}$ was used as the quantitative evaluation index. The visual detection results obtained from the proposed method on the original HSI and the noise-added HSI are shown in Fig. 13. The corresponding $\mathrm{AUC}_{(\mathrm{D,F})}$ performance comparison is shown in Table IX.

TABLE IX

$\mathrm{AUC}_{(\mathrm{D,F})}$ PERFORMANCE COMPARISON OF THE DETECTION RESULTS FROM THE PROPOSED METHOD ON THE ORIGINAL HSI AND THE HSI UNDER NOISE

Dataset	No noise	Gaussian noise	Salt and pepper noise
San Diego I	**0.9959**	0.9928	0.9871
San Diego II	**0.9978**	0.9977	0.9968
Pavia	0.9941	**0.9945**	0.9692
Texas Coast	0.9978	**0.9980**	0.9894
Salinas	**0.9999**	0.9998	0.9997
MUUFL Gulfport	**0.9923**	0.9785	0.8749

The best detection result was achieved for the input without noise as shown in Fig. 13. The background suppression effect was slightly worse under Gaussian noise. With salt and pepper noise, the background suppression effect was worse than with the original HSI and the HSI with Gaussian noise. Although the detection results from the proposed method were slightly worse under noise, it was still able to effectively detect the anomaly targets. According to Table IX, on the Pavia and Texas Coast datasets, although $\mathrm{AUC}_{(\mathrm{D,F})}$ values of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

TABLE X

$AUC_{(D,F)}$ PERFORMANCE COMPARISON OF DIFFERENT ANOMALY DETECTION METHODS ON THE SYNTHETIC AND THE MUUFL GULFPORT I DATASETS

Dataset	Method							
	RX	LRX	CRD	LRASR	AE	RGAE	Auto-AD	GAED
Synthetic dataset	0.4011	0.9991	0.9999	0.9650	0.9732	0.8289	0.9974	0.9954
MUUFL Gulfport I	0.9999	0.9999	0.9999	0.9999	0.9822	0.9841	0.9993	0.9997

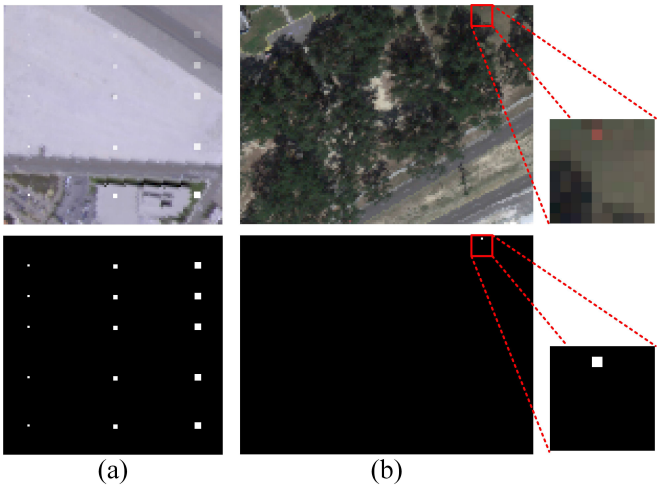

Fig. 14. Synthetic dataset and MUUFL Gulfport I dataset. (a) Synthetic dataset. (b) MUUFL Gulfport I dataset.

detection results under Gaussian noise to the input HSI were larger than those with the original HSI, the visual detection results and $AUC_{(D,F)}$ values remained very close. On other datasets, $AUC_{(D,F)}$ values of the detection results on the original HSI were larger than $AUC_{(D,F)}$ values of the detection results on HSI after adding noise. Therefore, the proposed method had some noise suppression properties.

G. Small Anomaly Targets' Detection

To verify whether the proposed method can detect small anomaly targets, we tested on a synthetic dataset and a real-world dataset.

1) Synthetic Dataset: The background of this dataset was collected by AVIRIS in San Diego, CA, USA [37] and it has a total of 100 × 100 pixels and 189 bands. The anomaly targets were embedded via the target implantation methods [51]. The target implantation method is written as

$$\boldsymbol{A}_{\mathrm{tg}} = \xi \times \boldsymbol{t}_{\mathrm{tg}} + (1 - \xi) \times \boldsymbol{b}_{\mathrm{bg}} \tag{13}$$

where $\boldsymbol{A}_{\mathrm{tg}}$ is the spectrum of the implanted anomaly target, ξ is the implanted abundance fraction, $\boldsymbol{t}_{\mathrm{tg}}$ is the spectrum of the implanted anomaly target, and $\boldsymbol{b}_{\mathrm{bg}}$ is the spectrum of the implanted background. A false-color image of the synthetic dataset is shown in Fig. 14(a) with 15 implanted anomaly targets with a total of 70 pixels. The anomaly target sizes in each column from left to right are 1 × 1, 2 × 2, and 3 × 3, respectively. The abundance fractions ξ for each row from top to bottom are 0.2, 0.4, 0.6, 0.8, and 1.0, respectively. The corresponding ground-truth image is shown in Fig. 14(a).

2) MUUFL Gulfport I: This dataset was cropped from the HSI collected by CASI–1500 at the University of Southern Mississippi Gulf Park Campus, Long Beach, MS, USA [49]. It has a total of 105 × 140 pixels and 64 bands. The cloth panel in the upper right corner is considered the anomaly target, which has one pixel. The false-color image and the corresponding ground-truth image of this dataset are shown in Fig. 14(b).

The detection results of different methods on the synthetic datasets are shown in Fig. 15. A visual inspection of the detection results indicated that the GRX method could not distinguish the anomalous targets from the background very well. The LRX method did not achieve satisfactory separation between the anomalous targets and the background. The LRASR, AE, and RGAE methods were marginally worse than the other detection results for the background suppression effect. Among the detection results of the CRD, Auto-AD, and the proposed GAED methods, the CRD method performed better than the rest. The corresponding $AUC_{(D,F)}$ performance comparisons are shown in Table X. The CRD method was found to achieve the best performance with $AUC_{(D,F)}$ value of 0.9999. Although the proposed GAED method was not best performing, its $AUC_{(D,F)}$ value reached 0.9954, which was close to that of the CRD method.

The detection results of different anomaly detection methods on the MUUFL Gulfport I datasets are shown in Fig. 16. Although all the methods detect anomaly targets from the visual detection results, the detection result of the proposed GAED method has the best background suppression effect. The anomaly targets in the detection results of the RX, the LRX, the CRD, and the LRASR methods are clearer, but there are also some false alarms in their detection results. The corresponding $AUC_{(D,F)}$ performance comparisons are shown in Table X. It can be seen from Table X that the RX, the LRX, the CRD, and the LRASR methods achieve the best detection performance. $AUC_{(D,F)}$ values of their test results are all 0.9999. $AUC_{(D,F)}$ value of the proposed GAED method also reaches 0.9997, which is very close to the best $AUC_{(D,F)}$ value of 0.9999.

The detection results of different anomaly detection methods on the MUUFL Gulfport I datasets are shown in Fig. 16. Visual inspection revealed that the proposed GAED method produced the best detection result for background suppression. The anomaly targets in the detection results of the RX, the LRX, the CRD, and the LRASR methods were clearer. However, some false alarms were found in the detection results. The corresponding $AUC_{(D,F)}$ performance comparisons are shown in Table X. The RX, the LRX, the CRD, and the LRASR methods achieved the best detection performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

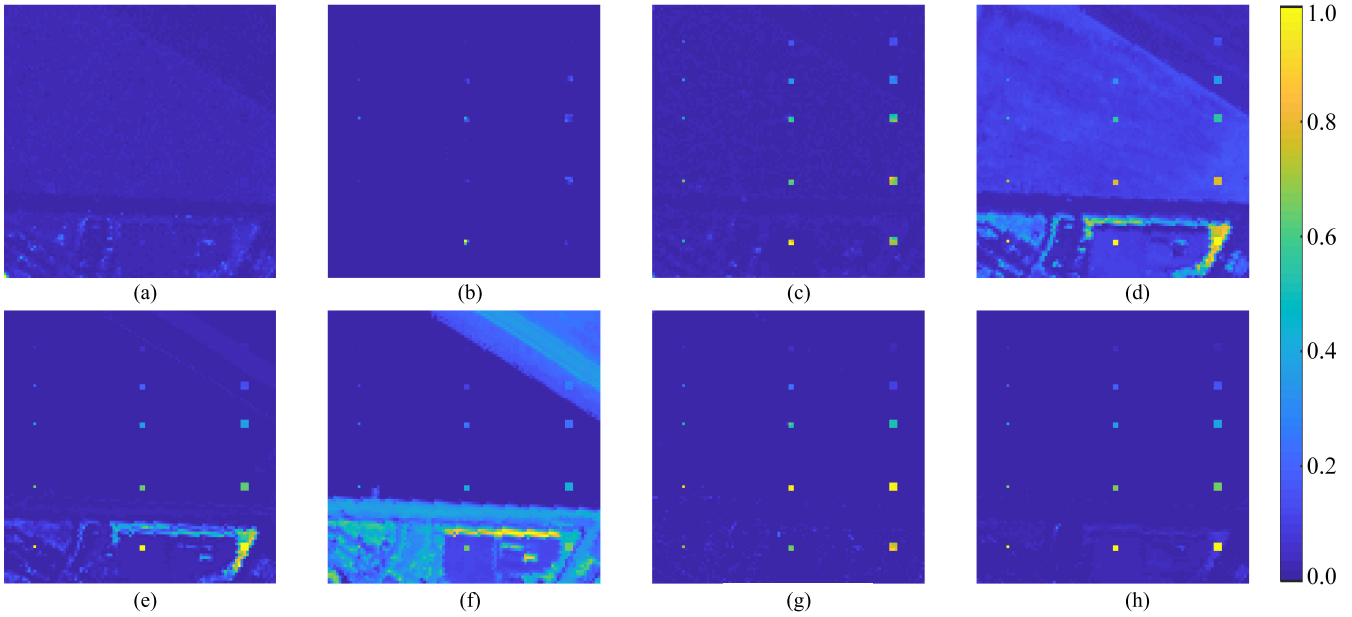

Fig. 15. Detection results of different anomaly detection methods on the synthetic dataset. (a) GRX. (b) LRX. (c) CRD. (d) LRASR. (e) AE. (f) RGAE. (g) Auto-AD. (h) GAED.

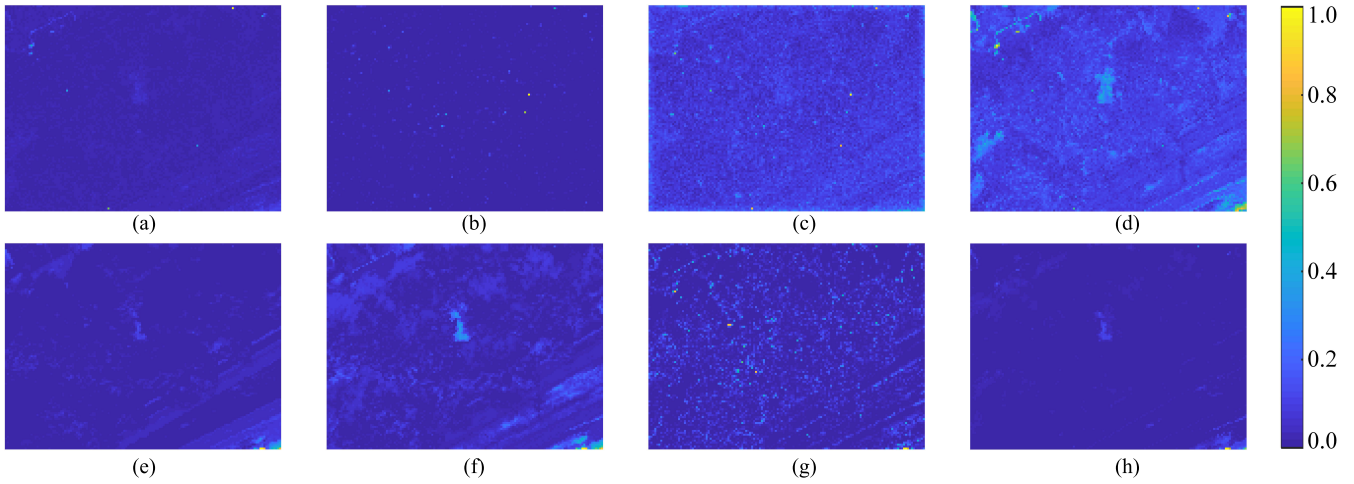

Fig. 16. Detection results of different anomaly detection methods on the MUUFL Gulfport I dataset. (a) GRX. (b) LRX. (c) CRD. (d) LRASR. (e) AE. (f) RGAE. (g) Auto-AD. (h) GAED.

$AUC_{(D,F)}$ values of their test results were all 0.9999. $AUC_{(D,F)}$ value of the proposed GAED method also reached 0.9997, which was very close to the best $AUC_{(D,F)}$ value of 0.9999. Even though the GAED method did not offer the best detection performance for small anomaly targets, its performance was very close to that of the best method.

V. Conclusion

This study proposes a novel guided AE-based hyperspectral anomaly detection method. The proposed method enhances the background reconstruction of AE and reduces the reconstruction of the anomaly targets of AE. This improves the anomaly detection performance. First, in the proposed method, we propose a multilayer AE with skip connections to improve the feature representation capability of the network. Second, a spectral-similarity-based method is proposed to obtain the guided image. This method uses the local spatial features of the HSI. Then, a guided module based on the obtained guided image is embedded into the backpropagation of the proposed AE network to enhance the feature representation of the background and reduce the feature representation of the anomaly targets by the proposed AE network. Finally, the final detection result can be obtained by the reconstructed error. In the experiment, the qualitative and quantitative comparisons between the proposed method and other state-of-the-art comparison methods were conducted on six datasets.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

The experimental results demonstrated the effectiveness and superiority of the proposed method.

ACKNOWLEDGMENT

The authors thank the relevant researchers for their datasets and codes.

REFERENCES

[1] T. Cheng and B. Wang, "Total variation and sparsity regularized decomposition model with union dictionary for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1472–1486, Feb. 2021.

[2] F. Xiong, J. Zhou, S. Tao, J. Lu, and Y. Qian, "SNMF-Net: Learning a deep alternating neural network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022, doi: 10.1109/TGRS.2021.3081177.

[3] M. Song, F. Li, C. Yu, and C.-I. Chang, "Sequential band fusion for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022, doi: 10.1109/TGRS.2021.3069734.

[4] L. Zhuang, L. Gao, B. Zhang, X. Fu, and J. M. Bioucas-Dias, "Hyperspectral image denoising and anomaly detection based on low-rank and sparse representations," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2021, doi: 10.1109/TGRS.2020.3040221.

[5] F. Xiong, J. Zhou, and Y. Qian, "Material based object tracking in hyperspectral videos," *IEEE Trans. Image Process.*, vol. 29, pp. 3719–3733, 2020.

[6] J. Qu, Q. Du, Y. Li, L. Tian, and H. Xia, "Anomaly detection in hyperspectral imagery based on Gaussian mixture model," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9504–9517, Nov. 2021.

[7] P. Xiang, J. Song, H. Qin, W. Tan, H. Li, and H. Zhou, "Visual attention and background subtraction with adaptive weight for hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2270–2283, Jan. 2021.

[8] J. Liu, Y. Feng, W. Liu, D. Orlando, and H. Li, "Training data assisted anomaly detection of multi-pixel targets in hyperspectral imagery," *IEEE Trans. Signal Process.*, vol. 68, pp. 3022–3032, 2020.

[9] H. Su, Z. Wu, H. Zhang, and Q. Du, "Hyperspectral anomaly detection: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 64–90, Mar. 2022.

[10] F. Li, X. Zhang, L. Zhang, D. Jiang, and Y. Zhang, "Exploiting structured sparsity for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 4050–4064, Jul. 2018.

[11] X. Ma, X. Zhang, X. Tang, H. Zhou, and J. Licheng, "Hyperspectral anomaly detection based on low-rank representation with data-driven projection and dictionary construction," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2226–2239, 2020.

[12] E. Bati, A. Çaliskan, A. Koz, and A. A. Alatan, "Hyperspectral anomaly detection method based on auto-encoder," *Proc. SPIE*, vol. 9643, Oct. 2015, Art. no. 96430N.

[13] G. E. Hinton and R. R. Salakhutdinov, "Reducing the dimensionality of data with neural networks," *Science*, vol. 313, no. 5786, pp. 504–507, 2006.

[14] W. Xie, B. Liu, Y. Li, J. Lei, and Q. Du, "Autoencoder and adversarial-learning-based semisupervised background estimation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5416–5427, Aug. 2020.

[15] G. Fan, Y. Ma, X. Mei, F. Fan, J. Huang, and J. Ma, "Hyperspectral anomaly detection with robust graph autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, doi: 10.1109/TGRS.2021.3097097.

[16] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, doi: 10.1109/TGRS.2021.3057721.

[17] S. I. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech Signal Process.*, vol. 38, no. 1, pp. 1760–1770, Nov. 1990.

[18] D. W. J. Stein, S. G. Beaven, L. E. Hoff, E. M. Winter, A. P. Schaum, and A. D. Stocker, "Anomaly detection from hyperspectral imagery," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 58–69, Jan. 2002.

[19] S. Matteoli, M. Diani, and G. Corsini, "Improved estimation of local background covariance matrix for anomaly detection in hyperspectral images," *Opt. Eng.*, vol. 49, no. 4, p. 258, Apr. 2010.

[20] A. P. Schaum, "Hyperspectral anomaly detection: Beyond RX," *Proc. SPIE*, vol. 6565, May 2007, Art. no. 656502.

[21] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, and A. Plaza, "Weighted-RXD and linear filter-based RXD: Improving background statistics estimation for anomaly detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014.

[22] S. Chen, C.-I. Chang, and X. Li, "Component decomposition analysis for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–22, 2022, doi: 10.1109/TGRS.2021.3117765.

[23] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[24] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Sparse representation for target detection in hyperspectral imagery," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 629–640, Jun. 2011.

[25] L. Wei and D. Qian, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[26] S. Li, W. Wang, H. Qi, B. Ayhan, C. Kwan, and S. Vance, "Low-rank tensor decomposition based anomaly detection for hyperspectral imagery," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2015, pp. 4525–4529.

[27] W. Sun, C. Liu, J. Li, Y. M. Lai, and W. Li, "Low-rank and sparse matrix decomposition-based anomaly detection for hyperspectral imagery," *J. Appl. Remote Sens.*, vol. 8, no. 1, May 2014, Art. no. 083641.

[28] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.

[29] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[30] Y. Qu *et al.*, "Hyperspectral anomaly detection through spectral unmixing and dictionary-based low-rank decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4391–4405, Aug. 2018.

[31] L. Li, W. Li, Q. Du, and R. Tao, "Low-rank and sparse decomposition with mixture of Gaussian for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4363–4372, Sep. 2021.

[32] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, and Q. Du, "Prior-based tensor approximation for anomaly detection in hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1037–1050, Mar. 2022.

[33] T. Cheng and B. Wang, "Graph and total variation regularized low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 391–406, Jan. 2020.

[34] S. Chang, B. Du, and L. Zhang, "A sparse autoencoder based hyperspectral anomaly detection algorihtm using residual of reconstruction error," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 5488–5491.

[35] W. Xie, J. Lei, B. Liu, Y. Li, and X. Jia, "Spectral constraint adversarial autoencoders approach to feature representation in hyperspectral anomaly detection," *Neural Netw.*, vol. 119, pp. 222–234, Nov. 2019.

[36] C. Zhao and L. Zhang, "Spectral–spatial stacked autoencoders based on low-rank and sparse matrix decomposition for hyperspectral anomaly detection," *Infr. Phys. Technol.*, vol. 92, pp. 166–176, Aug. 2018.

[37] S. Arisoy, N. M. Nasrabadi, and K. Kayabol, "Unsupervised pixel-wise hyperspectral anomaly detection via autoencoding adversarial networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3049711.

[38] X. Lu, W. Zhang, and J. Huang, "Exploiting embedding manifold of autoencoders for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1527–1537, Mar. 2020.

[39] A. Creswell and A. A. Bharath, "Denoising adversarial autoencoders," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 4, pp. 968–984, Apr. 2019.

[40] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1026–1034.

[41] P. Xiang *et al.*, "Hyperspectral anomaly detection by local joint subspace process and support vector machine," *Int. J. Remote Sens.*, vol. 41, no. 10, pp. 3798–3819, Jan. 2020.

[42] Y. Gong and I. F. Sbalzarini, "Curvature filters efficiently reduce certain variational energies," *IEEE Trans. Image Process.*, vol. 26, no. 4, pp. 1786–1798, Apr. 2017.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.

[43] W. Xie, B. Liu, Y. Li, J. Lei, and Q. Du, "Autoencoder and adversarial-learning-based semisupervised background estimation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5416–5427, Aug. 2020.
[44] F. Xiong, J. Zhou, Q. Zhao, J. Lu, and Y. Qian, "MAC-Net: Model-aided nonlocal neural network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, doi: 10.1109/TGRS.2021.3131878.
[45] Y. Wu *et al.*, "Direct training for spiking neural networks: Faster, larger, better," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, no. 1, Jul. 2019, pp. 1311–1318.
[46] X. Lu, W. Zhang, and X. Li, "A hybrid sparsity and distance-based discrimination detector for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1704–1717, Dec. 2018.
[47] X. Kang, X. Zhang, S. Li, K. Li, J. Li, and J. A. Benediktsson, "Hyperspectral anomaly detection with attribute and edge-preserving filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5600–5611, Oct. 2017.
[48] F. Verdoja and M. Grangetto, "Graph Laplacian for image anomaly detection," *Mach. Vis. Appl.*, vol. 31, nos. 1–2, pp. 1432–1769, Feb. 2020.
[49] X. Du and A. Zare, "Technical report: Scene label ground truth map for MUUFL Gulfport data set," Univ. Florida, Gainesville, FL, USA, Tech. Rep. 20170417, Apr. 2017.
[50] C.-I. Chang, "An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis," *IEEE Trans. Geosci. Remote Sensing*, vol. 59, no. 6, pp. 5131–5153, Jun. 2021.
[51] M. S. Stefanou and J. P. Kerekes, "A method for assessing spectral image utility," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 6, pp. 1698–1706, Jun. 2009.

Pei Xiang received the B.S. degree in electrical engineering and automation from Qinghai University, Xining, China, in 2016. He is currently pursuing the Ph.D. degree in electronic science and technology with the School of Physics and Optoelectronic Engineering, Xidian University, Xi'an, China. He is also a joint Ph.D. Student with the School of Computer Science and Engineering, Kyungpook National University, Daegu, South Korea.

His research interests include hyperspectral anomaly detection, remote sensing image processing, and machine learning.

Shahzad Ali received the B.S. degree in computer and information sciences from the Pakistan Institute of Engineering and Applied Sciences, Islamabad, Pakistan, in 2012, and the M.S. degree in computer science from the National University of Computer and Emerging Science, Lahore, Pakistan, in 2016. He is currently pursuing the Ph.D. degree with the Virtual Reality Laboratory, Kyungpook National University, Daegu, South Korea.

His research interests include span computer vision, machine learning, biomedical image processing, and remote sensing.

Soon Ki Jung (Senior Member, IEEE) received the M.S. and Ph.D. degrees in computer science from the Korea Advanced Institute of Science and Technology, Daejeon, South Korea, in 1992 and 1997, respectively.

He was a Visiting Professor with the University of Southern California, Los Angeles, CA, USA, in 2002 and 2009. He is currently the Head of the Graduate School of Computer Science and Engineering, Kyungpook National University, Daegu, South Korea. His research areas include a broad range of computer vision, computer graphics, and virtual reality topics.

Dr. Jung has been an active Executive Board Member of Human–Computer Interaction, Computer Graphics, and Multimedia societies in South Korea.

Huixin Zhou (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in optical engineering from Xidian University, Xi'an, China, in 1996, 2002, and 2004, respectively.

He is currently a Professor with the School of Physics and Optoelectronic Engineering, Xidian University. His research interests include remote sensing image processing, photoelectric imaging, real-time image processing, and target detection and tracking.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:39:37 UTC from IEEE Xplore. Restrictions apply.